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Design and test of a CMOS folded cascode OTA with current recycling

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Acronyms

AC Alternating Current

CMOS Complementary Metal-Oxide-Semiconductor

CMRR Common-Mode Rejection Ratio

DC Direct Current

DFT Discrete Fourier Transform

DRC Design Rules Check

GBW Gain Bandwidth Product

HD Harmonic Distortion

ICMR Input Common Mode Range

LVS Layout vs Schematic

MOS Metal-Oxide-Semiconductor

NMOS N-Channel Metal-Oxide-Semiconductor

OTA Operational Transconductance Amplifier

PDK Process Design Kit

PEX Parasitic Extraction

PMOS P-Channel Metal-Oxide-Semiconductor

PN P-Type and N-Type

PSRR Power Supply Rejection Ratio

RMS Root-Mean-Square

RVE Results Viewing Environment

VAT Value-Added Tax

VPN Virtual Private Network

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Abstract

This thesis covers the design of a Complementary Metal-Oxide-Semiconductor (CMOS) folded Cascode Operational Transconductance Amplifier (OTA) in UMC 180nm technology. The broad use of operational transconductance amplifiers in the industry requires them to be highly efficient to improve the operation of the circuits they are included in.

A technique called current recycling is known to improve the power efficiency of the amplifier with a minimal complication of the circuit. This thesis covers the standard implementation of the folded cascode amplifier and the current recycling folded cascode and simulations to compare them in terms of performance, checking both pros and cons of both implementations. Layout and Post Layout simulations of the current recycling folded cascode amplifier will also be carried out.

Furthermore, a biasing circuit has also been included in both designs, which will provide the bias voltage without the need of extra sources other than V_{DD}. The implemented design of the biasing circuit is a design that it is insensitive to power source fluctuations based on a stable transconductance matched to the conductance of a resistor.

1 Introduction

Operational Transconductance Amplifiers (OTA) is a common and widely used amplifying device used in voltage or current mode devices. One of the most popular structures for the OTA is a folded cascode structure, that is widely chosen because of the high output impedace, low noise characteristic and wide output swing.

The continuous integration of complex analog and digital building blocks on a single chip make silicon area and power consumption two very valued aspects when designing integrated circuits. Even having all those previous pros and despite being one of the most important blocks in Analog Design and vital in many applications, in the mobile and portable industry the OTA is still the largest component and also the block that usually has the largest power consumption, which limits the usage of this device in this field [1].

Different approaches have been attempted to reduce both size and power consumption, but in this thesis, the one presented is Current Recycling [3]. Previous work tried multi-path schemes [11] to enhance output impedance and slew rate, nonetheless, the main drawback of that approach is that it made the OTA not suitable for high speed applications due to the low frequency pole-zero pairs[3].

Current recycling theoretical study shows that with similar area and same power budget it can boost the gain and slew-rate without adversely affecting the input referred noise and the input offset [3] . This thesis goes through different checks to verify the performance boost and improvements and the adverse effects of using this structure is minimal and it is a structure worth to consider when designing an OTA.

1.1 Methodology and procedures

For the development of this thesis, both Conventional and Recycling current structure are compared. Both amplifiers are designed in schematics level in Cadence Virtuoso and then, the physical design verification is done through Calibre.

Theoretical simulations are performed for both structures and then compared to see which structure has better performance and also to verify and check any limitations on the proposed Recycling structure. Other parameters are also checked to verify that no other performance metric is affected adversely.

Layout is performed on the Current Recycling structure. Post Layout simulations are also performed to check and get a closer performance simulation of the physical block of the design.

1.2 Scope of work

Due to time limitation constrains, this work does not include the tape-out of the chip and the experimental verification of the design.

This work focuses on the implementation and the layout and the simulations from Cadence Virtuoso.

1.3 Work requirements and specifications

The following requirements and specifications are considered for the development of this thesis:

1.3.1 Requirements

1. Recycling Folded Cascode Gain Bandwidth Product (GBW): $150MHz$.
2. Capacitive load (C_L): 2.25pF
3. For a fair comparison in the performance, both amplifiers must have similar current consumption to demonstrate that, with the same current consumption, the performance of the Recycling structure is better than the Conventional structure.

1.3.2 Specifications

1. Supply voltage: 1.8V
2. Use same test bench for comparison
3. Use the same biasing circuit to generate the bias voltage of both circuits

1.4 Gantt Diagram

The timing of this thesis is divided in two big blocks: schematic and layout. Besides these two blocks, an initial part of the project was to get familiar with the setup, specially with the physical validation in Calibre. The initial project timing was projected to be as follows:

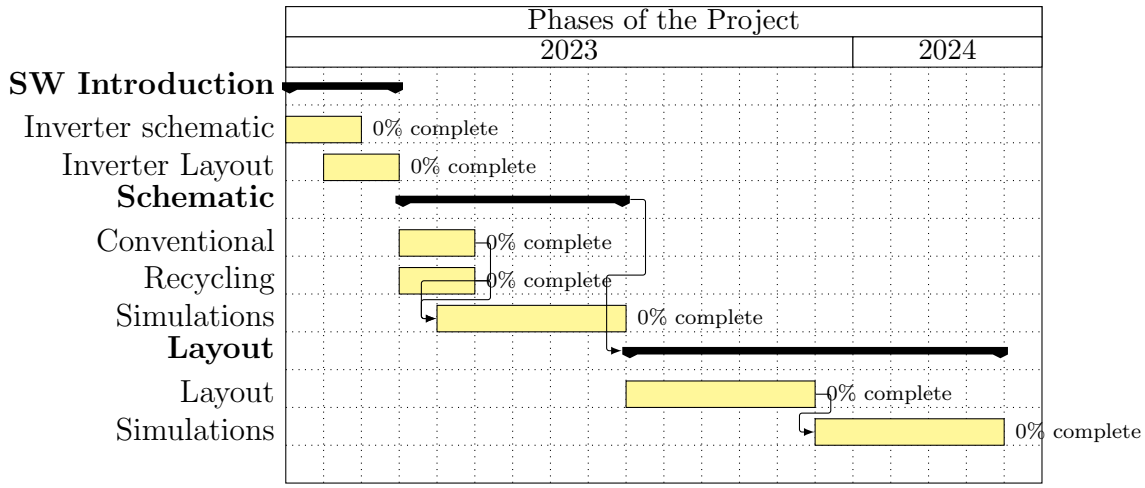


Figure 1: Initial Gantt diagram of the project

The final timing of the project is as follows:

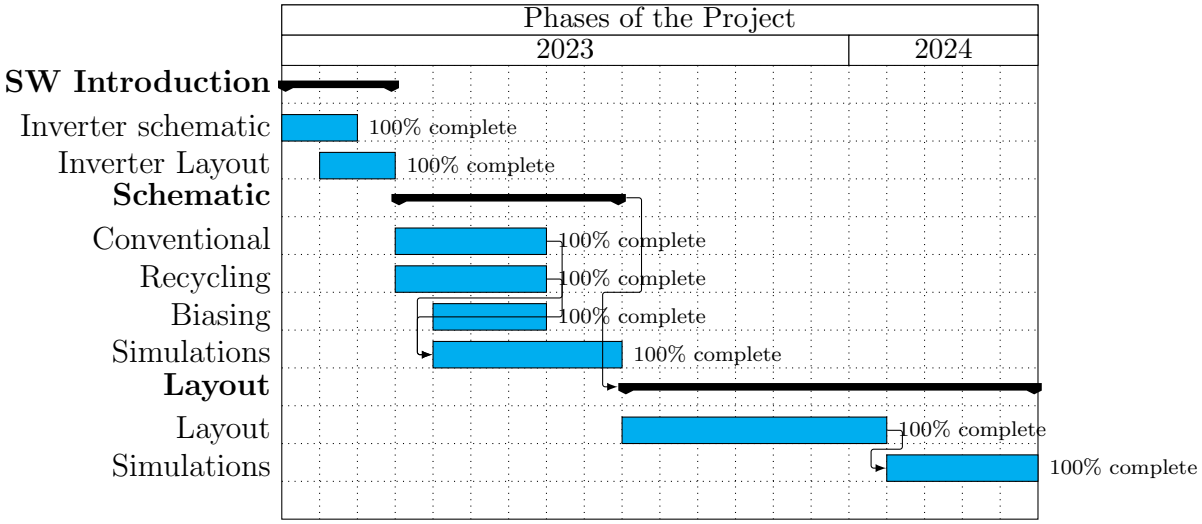


Figure 2: Final Gantt diagram of the project

The differences in the Gantt diagrams are briefly explained:

1. Longer time in schematic design:

   1.1 Delay due to biasing circuit. The initial design of the amplifiers did not include a biasing circuit, the biasing of the amplifiers was implemented using other

methodologies that required external current sources. During the development of the thesis, it was decided that a biasing circuit should be included to avoid the usage of extra external sources other than V_{DD} in case of implementing on a chip. Furthermore, a biasing circuit that is insensitive to power source fluctuations and has a constant transconductance is selected, for a more stable system against external variations.

1.2 Delay in the sizing of the amplifiers. From initial simulations, the Recycling Folded Cascode did not reach enough Phase Margin to consider the system stable, which lead to iterations in the sizing and the capacitance value. More detailed explanation in section 5.

2. Delay in layout

2.1 There was a delay in the layout due to Parasitic Extraction (PEX) not working correctly. This lead to multiple tests and debugging to try to find the error and also reinstallation of Cadence configuration.

2.2 Layout iteration. Initial proposed layout for the input transistors used a common-centroid scheme in a 4x4 shape. This made the common-centroid to be in the middle and all the transistors had the same distance to the center. Nonetheless, the main drawback of that design is in terms of sizing. The input transistors are the wider, and having them in a 4x4 shape, made the layout larger because of the space that the rows in the 4x4 structure needed in between to allow interconnection. By changing to a 16x1 structure and applying interdigitation a more compact solution is achieved. Furthermore, given that interdigitation is applied, this does not affect negatively the matching between transistors, since the common-centroid structure is mantained.

2 Conventional Folded Cascode

2.1 Structure

The structure of the conventional Folded Cascode OTA is depicted in Figure 3. This amplifier is widely used because of its high gain and large signal swing. The most common structure is composed by P-Channel Metal-Oxide-Semiconductor (PMOS) input, given that it provides a higher non-dominant pole, lower flicker noise and input common mode level[16].

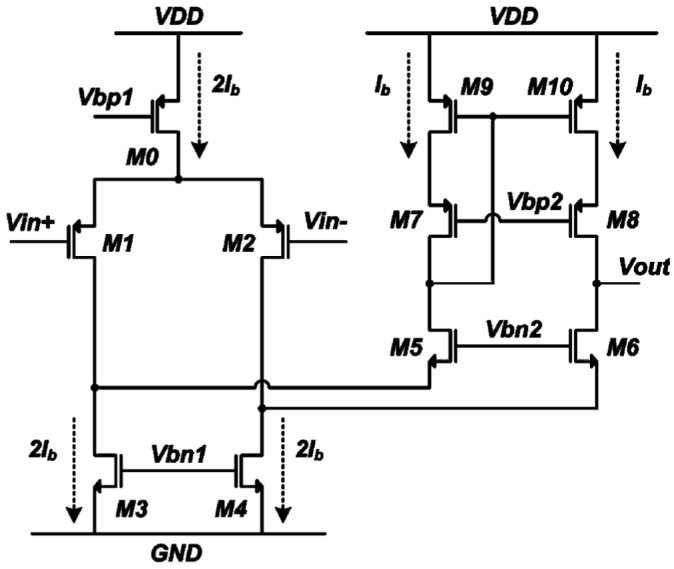


Figure 3: Conventional Folded Cascode Structure [3]

Despite the previous presented advantages and being one of the most used blocks in analog design, one of the drawbacks of this design is the inefficiency in transistors *M3* and *M4*. Circuit analysis shows that these two transistors are the transistors that conduct the highest current in the circuit, thus, in the majority of designs are the transistors with highest transconductance. This is highly inefficient if the role of these transistors is considered: both transistors are only used to provide a folding node for the small signal current generated by the input transistors.

The amplifier is a single stage with differential input and single ended-output. The input transistors are formed by *M1* and *M2*. To the drain of transistors *M3* and *M4* there are also connected the cascode transistors *M5* and *M6* and, to do the differential-to-single-ended conversion, transistors *M7* to *M10* are implementing a wide-swing current mirror [9].

What makes this structure one of the most widely used is that with the size and the simplicity of a single-stage amplifier it can reach the performance of a multi-stage amplifier because of the usage of cascode structures in the output stage, which make the output impedance high.

The biasing circuit specified in section 4 generates the biasing voltages in figure 3.

2.2 Characteristics

All the transistors shown in the circuit and in all the following sections are working in saturation region and the simplified square-law drain current model is used to compute an initial approximation and to develop and show the characteristics of the amplifier.

$$I_D = \frac{1}{2} \cdot \mu \cdot C_{ox} \cdot \frac{W}{L} \cdot (V_{GS} - V_T)^2 \tag{1}$$

The main characteristics that are looked into this section for further comparison are the transconductance, the gain and the slew rate, since these parameters are easily obtained and give a good overview of the amplifier performance.

Regarding the transconductance of the amplifier (Gm_{FC}) it is fixed by the transconductance of the input transistor (gm_1), thus:

$$Gm_{FC} = gm_1 \tag{2}$$

For the gain ($A_{v_{FC}}$), the expression is:

$$A_{v_{FC}} = Gm_{FC} \cdot R_{out_{FC}} \tag{3}$$

Where Gm_{FC} is the transconductance of the amplifier and $R_{out_{FC}}$ is the output impedance of the amplifier.

By analysing the circuit, the output impedance considering the parasitic resistors of the active load is expressed as:

$$R_{out_{FC}} = gm_6 \cdot r_{ds6} \cdot (r_{ds2} || r_{ds4}) || gm_8 \cdot r_{ds8} \cdot r_{ds10} \tag{4}$$

Where gm_i is the transconductance of transistor i and r_{dsi} is the drain-source resistor of transistor i.

Therefore, by substituting in the gain expression, the resulting formula for the gain is:

$$A_v = gm_1 \cdot gm_6 \cdot r_{ds6} \cdot (r_{ds2}||r_{ds4})||gm_8 \cdot r_{ds8} \cdot r_{ds10} \tag{5}$$

The expression for the slew rate (SR) is given by:

$$SR = \frac{2 \cdot I_b}{C_L} \tag{6}$$

In that expression two factors take into account: the maximum current that flows through the loads ($2 \cdot I_b$, where I_b is the current defined in figure 3) and the capacitive load (C_L).

2.3 Sizing

For the sizing of this amplifier, a first approximation is performed to obtain the first Width and Length ratios. The followed steps to obtain this initial approximation are:

1. Obtain the transconductance for the input transistors

   The transconductance of the input transistors (gm_1) are obtained with the Gain-Bandwidth product (GBW) and the load capacitance (C_L). From the specifications of the thesis, the Recycling Folded Cascode GBW has to be 150MHz. From [3], the GBW of the Recycling structure is twice as large as the conventional. Thus, for the sizing of the conventional a GBW of 100MHz is used, to have some margin due to the usage of the square-law drain current model and further approximations in the sizing. The C_L value is also fixed from the specifications and it is 2.25pF.

   This value is an approximation of what ideally the transistor should have if all the components are ideal. The expression used is:

$$gm_1 = 2 \cdot \pi \cdot C_L \cdot GBW \tag{7}$$

2. Obtain the $\frac{W}{L}$ ratio for the input transistors

   With the previously obtained transconductance, calculate the sizing ratio required for the input transistors:

$$gm_1 = K_P \cdot \frac{W}{L} \cdot (V_{GS} - V_T) \tag{8}$$

$$\frac{W}{L} = \frac{gm}{K_P \cdot (V_{GS} - V_T)} \tag{9}$$

   Where K_P is $K_P = \mu_p \cdot C_{ox}$ and $(V_{GS} - V_T)$ is the overdrive voltage, that is defined by the gate-source voltage minus the threshold voltage.

3. Fix an overdrive voltage

   As seen in the expression there is a parameter that is unknown and it is a degree of freedom in the design, which is the overdrive voltage $(V_{GS} - V_T)$. For the initial approximation a value of $150mV$ is selected for all transistors. This value is the mean value from two different methodologies to fix the overdrive voltage. The first methodology is to fix an overdrive voltage of $200mV$[13], and the other methodology is to use the 20% of V_{DD}[4].

   For the value of K_P, a first approximation was done by simulations. The value used for the calculations is 63 $\mu A/V^2$.

4. Fix the length of the input transistors

   To reduce the channel modulation effects, the length of most of the transistors is $3 \cdot MinimumLength$. In this case, the selected value of length for the input transistors is $540nm$, given that the minimum length for this technology is $180nm$.

5. Compute the Width of the input transistors

   With the aspect ratio and the length that is fixed in the previous step, the width of the input transistors are obtained.

6. Calculate the current through input transistors

   With the width and length values fixed and using the simplified square-law drain current model:

$$I_D = \frac{1}{2} \cdot K_P \cdot \frac{W}{L} \cdot (V_{GS} - V_T)^2 \tag{10}$$

7. Calculate the current passing through all the other transistors

   When the current passing through the input transistor is known, from the schematic in figure 3, all the currents flowing through the different transistors in the circuit are calculated.

8. Calculate the $\frac{W}{L}$ for the rest of transistors

   For the calculation of the size ratios of the transistors the simplified square-law drain current model is used. The expressions that are given are for PMOS, for NMOS transistors, instead of the previously obtained K_P, an approximation of K_N is used. It was found using the same procedure and the value obtained was 280 $\mu A/V^2$:

$$I_D = \frac{1}{2} \cdot K_P \cdot \frac{W}{L} \cdot (V_{GS} - V_T)^2 \tag{11}$$

$$\frac{W}{L} = \frac{2 \cdot I_D}{K_P \cdot (V_{GS} - V_T)^2} \tag{12}$$

   Analogous expression used for NMOS:

$$\frac{W}{L} = \frac{2 \cdot I_D}{K_N \cdot (V_{GS} - V_T)^2} \tag{13}$$

   From previously given expression, the size ratio is obtained with the overdrive voltage specified and the approximation of K_P and K_N also given in previous steps.

9. Fix the Lengths of the transistors and obtain the Width

   For all the transistors, the same criteria as the input transistors is followed: $3 \cdot MinimumLength$. The only exception in the design are transistors *M9* and *M10*. Both of these transistors, since are part of the active load and the source is connected to V_{DD}, the channel length modulation effect is reduced, so a smaller length is used for these transistors, in this case only a length of $2 \cdot MinimumLength$ is selected for both transistors.

By following these steps, the resulting table summarizes the values obtained with the approximations done:

Recycling Folded Cascode Sizing [μm]		
M0	W	161.57
	L	0.54
	W/L	299.20
M1and M2	W	80.78
	L	0.54
	W/L	149.59
M3 and M4	W	36.36
	L	0.54
	W/L	67.32
M5 and M6	W	18.17
	L	0.54
	W/L	33.66
M7 and M8	W	80.78
	L	0.54
	W/L	149.59
M9 and M10	W	53.85
	L	0.36
	W/L	149.59

Table 1: Sizing of the transistors in the Conventional Folded Cascode

The resulting schematic is:

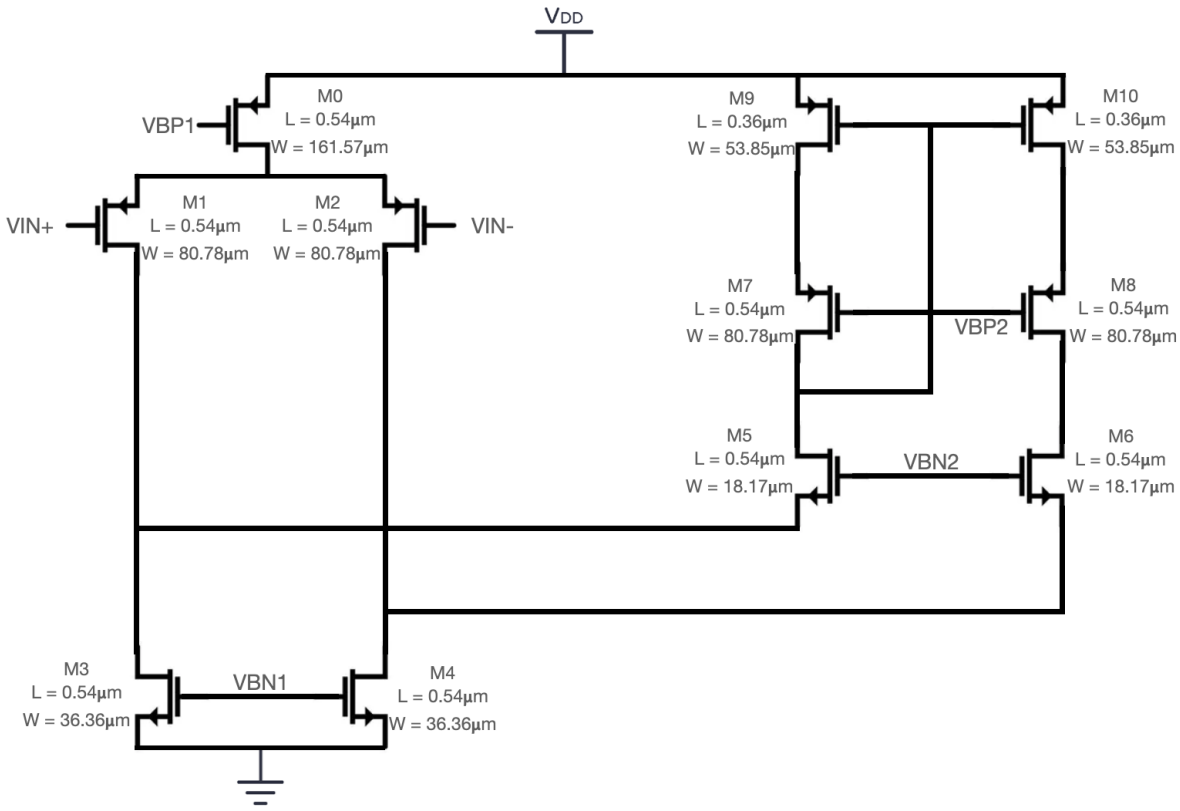


Figure 4: Conventional Folded Cascode Schematic with computed dimensions

3 Current Recycling Folded Cascode

3.1 Structure

The main structure of this amplifier is really similar to the structure of the Conventional Folded Cascode, but with some modifications to address the main drawback that the conventional structure has: the inefficiency of transistors *M3* and *M4*.

The modifications performed are to make *M3* and *M4* driving transistors. To do so, the input transistors *M1* and *M2* are divided in half. With the circuit structure depicted in figure 5, each of the half transistors drive half of the current that *M1* and *M2* were driving in the Conventional structure.

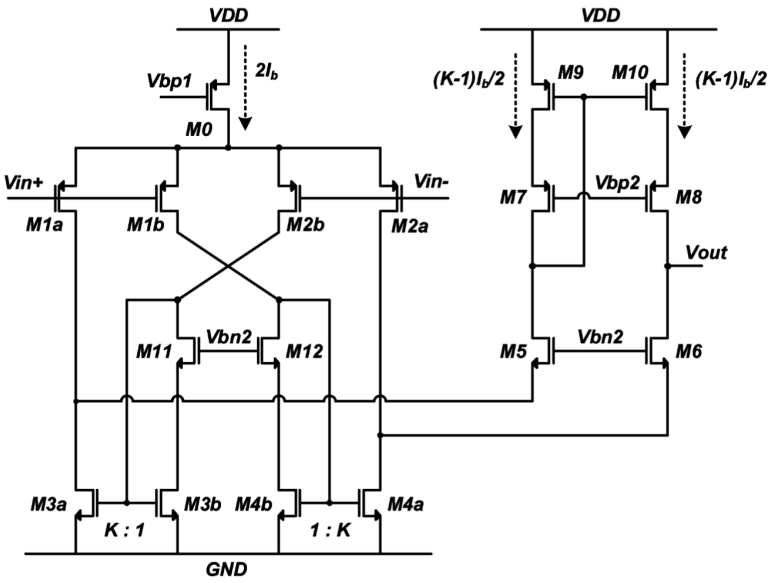


Figure 5: Recycling Folded Cascode Structure [3]

Another modification done is to the transistors that are intended to be modified: *M3* and *M4*. These transistors are split into current mirrors with a ratio of K:1. When the selected K is 3 ($K = 3$), as seen in the annotations in the circuits figure, the current consumption is the same as in the Conventional Folded Cascode, so, for a fair comparison, that value of K is selected. For another value of K, the load section must be resized to maintain the power consumption[3].

Two additional transistors are added in the structure, which are connected in between the input transistors and the smaller size branch of the current mirrors. These transistors are added to ensure that the small signal current added in the sources of *M5* and *M6* are in phase.

As observed, the circuit and structure modifications are relatively easy to perform and as shown in next section, from theoretical analysis, are supposed to deliver a higher performance.

3.2 Characteristics

The same characteristics as the Conventional Folded Cascode are analyzed: transconductace, gain and slew rate.

When it comes to the transconductace of the amplifier (Gm_{RFC}), the expression obtained for this amplifier is:

$$Gm_{RFC} = gm_{1A} \cdot (1 + K) \tag{14}$$

Where gm_{1A} is the transconductance of the input amplifier and the K is the design parameter previously selected as $K = 3$.

Regarding the gain, the expression keeps being the same, the amplifier transconductance (Gm_{RFC})multiplied with the output resistance ($R_{out_{RFC}}$), however, due to the changes mentioned, there are changes in the output resistance:

$$A_v = Gm_{RFC} \cdot R_{out_{RFC}} \tag{15}$$

$$R_{out_{RFC}} = gm_6 \cdot r_{ds6} \cdot (r_{ds2a}||r_{ds4a})||gm_8 \cdot r_{ds8} \cdot r_{ds10} \tag{16}$$

$$A_v = gm_{1A} \cdot (1 + K) \cdot gm_6 \cdot r_{ds6} \cdot (r_{ds2a}||r_{ds4a})||gm_8 \cdot r_{ds8} \cdot r_{ds10} \tag{17}$$

As in previous sections, gm_i is the transconductance of transistor i and r_{dsi} is the drain-source resistor of transistor i.

As observed, the modifications in this expression is that instead of considering the drain-source resistance of the input transistor ($M2$) and transistor $M4$, since both of them have been split in half, only the resistance of half transistor is considered. In the next section the effect of this slight change is discussed.

The slew rate in this case it is observed that in this structure, the current is higher because it is multiplied by K, which leads to a higher Slew Rate:

$$SR = \frac{2 \cdot K \cdot I_b}{C_L} \tag{18}$$

3.3 Comparison against Conventional Folded Cascode

From previous sections, a comparison of the characteristics between the Conventional Folded Cascode and the Recycling Folded Cascode is done.

For the transconductance of the amplifier, for the conventional structure the expression was:

$$Gm_{FC} = gm_1 \tag{19}$$

And for the Recycling Folded Cascode:

$$Gm_{RFC} = gm_{1a} \cdot (1 + K) \tag{20}$$

In the Conventional Folded Cascode, the input transistors are not split, thus, the size is doubled. Since the rest of the parameters remain equal, the current that the input transistors drive in the Conventional Folded Cascode is twice as large as in the Recycling, which leads also to twice the transconductance in the input transistors: $gm_1 = 2 \cdot gm_{1a}$

For the Recycling Folded Cascode, by substituting K, the expressions for the transconductance simplifies to:

$$Gm_{RFC} = gm_{1a} \cdot (1 + K) = 4 \cdot gm_{1a} \tag{21}$$

Then, the transconductance of the Conventional Folded Cascode expressed in terms of gm_{1a} :

$$Gm_{FC} = 2 \cdot gm_{1a} \tag{22}$$

From above expressions, it is seen that the transconductance of the Recycling structure is the double, which means with the same power consumption, the GBW of the Recycling is twice as large as the Conventional (considering that C_L is not changed):

$$GBW = 2 \cdot \pi \cdot C_L \cdot Gm \tag{23}$$

Another improvement that appears by changing to a Recycling structure is in the gain. From the increased transconductance (2 times larger), there is already a 6dB improvement in the Gain. Nonetheless, in the output resistance there was a change, given that another resistances need to be considered.

In the Recycling structure, half transistors are used to get the output resistance. Since these half transistors are driving less current, the value of those parasitic resistors increases, thus, increasing the overall output resistance.

From previous analysis, it was already observed that with the Recycling structure, the Slew Rate was higher, because the maximum current flowing through the load is higher.

Nonetheless, not all the parameters improve. With the changes in the structure, pole-zero pairs are added, affecting the Phase Margin. Both structures share the dominant pole in the output node, and also share the non-dominant pole due to the non-parasitics of the source of *M5* and *M6*.

In the Recycling structure, splitting *M3* and *M4* in current mirrors, generates a new pole-zero pair. Because the devices that generate the pole-zero pair are NMOS, the pair is located at high-frequency, but it is expected to worsen the Phase Margin in the Recycling compared to the Conventional.

3.4 Sizing

Since a fair comparison must be done between both amplifiers, the sizing of the Recycling structure is based on the sizing previously done for the Conventional Structure. The modifications presented in the Structure section are applied to the sizing obtained in the Conventional structure to obtain the sizing of the Recycling structure. The following list of changes shows the difference compared to the Conventional Folded Cascoded:

1. Input transistors split in half

2. *M3* and *M4* split into a 3:1 current mirror structure

To size the current mirror transistors, for the branch that drives the smaller current, the width is $\frac{1}{4}$ of the width of transistors *M3* and *M4* in the Conventional Structure.

For the branch that drives more current, the ratio of $K = 3$ is used, thus, the width is $\frac{3}{4}$ of the width of transistors *M3* and *M4* in the Conventional Structure.

3. Newly added transistors *M11* and *M12*

   For the sizing of both of these transistors, the simplified square-law drain current model is used. An alternative to calculate the sizing of these transistors is to use the sizing of transistors *M5* and *M6*. Given that the current flowing in *M11* and *M12* is the half of the current that flows through *M5* and *M6*, to have the same overdrive voltage, the size must be the half.

The table summarizing the sizes of this amplifier is:

Recycling Folded Cascode Sizing [μm]		
M0	W	161.57
	L	0.54
	W/L	299.20
M1a and M1b	W	40.39
	L	0.54
	W/L	74.80
M2a and M2b	W	40.39
	L	0.54
	W/L	74.80
M3a and M4a	W	27.24
	L	0.54
	W/L	50.4
M3b and M4b	W	9.09
	L	0.54
	W/L	16.83
M5 and M6	W	18.17
	L	0.54
	W/L	33.66
M7 and M8	W	80.78
	L	0.54
	W/L	149.59
M9 and M10	W	53.85
	L	0.36
	W/L	149.59
M11 and M12	W	9.09
	L	0.54
	W/L	16.83

Table 2: Sizing of the transistors in the Recycling Folded Cascode

The resulting schematic is:

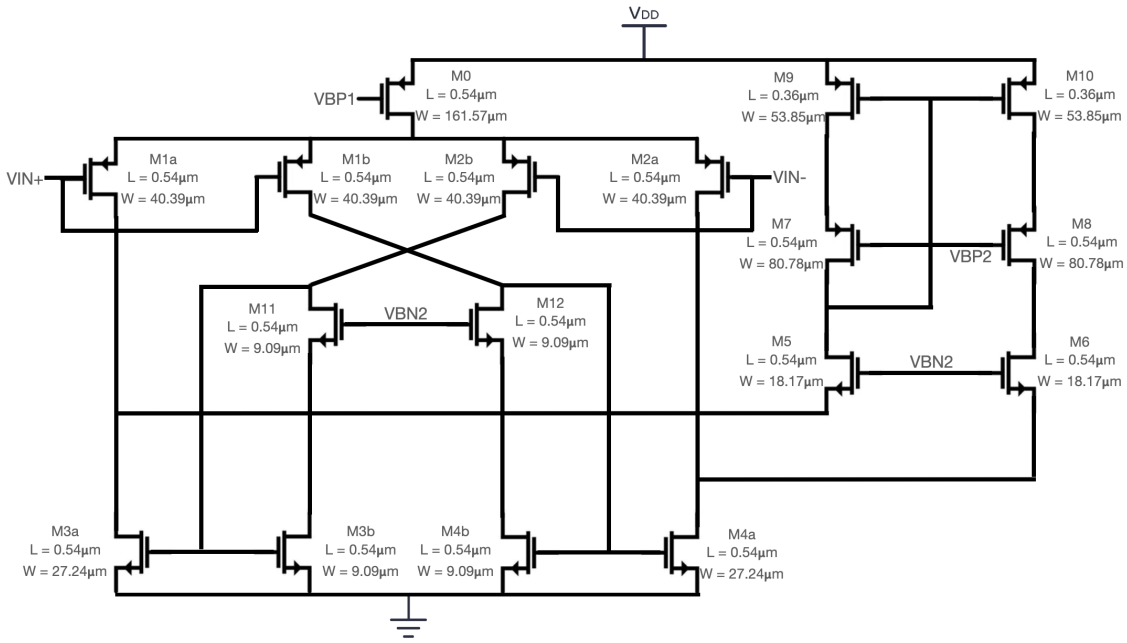


Figure 6: Recycling Folded Cascode Schematic with computed dimensions

The comparison table between both designs is table 3. The gate area slightly increases due to the addition of two transistors (*M11* and *M12*) and the total area increases too, because the transistors are split, nonetheless, the current consumption does not increase. If the area is not a limitation, from theoretical comparison of both options, the Recycling structure outperforms the Conventional structure.

Transistor		Conventional Structure [μm]	Recycling Structure [μm]
M0	W	161.57	161.57
	L	0.54	0.54
	W/L	299.20	299.20
M1 and M2	W	80.78	-
	L	0.54	-
	W/L	149.59	-
M1a and M1b	W	-	40.39
	L	-	0.54
	W/L	-	74.80
M2a and M2b	W	-	40.39
	L	-	0.54
	W/L	-	74.80
M3 and M4	W	36.36	-
	L	0.54	-
	W/L	67.32	-
M3a and M4a	W	-	27.24
	L	-	0.54
	W/L	-	50.4
M3b and M4b	W	-	9.09
	L	-	0.54
	W/L	-	16.83
M5 and M6	W	18.17	18.17
	L	0.54	0.54
	W/L	33.66	33.66
M7 and M8	W	80.78	80.78
	L	0.54	0.54
	W/L	149.59	149.59
M9 and M10	W	53.85	53.85
	L	0.36	0.36
	W/L	149.59	149.59
M11 and M12	W	-	9.09
	L	-	0.54
	W/L	-	16.83

Table 3: Sizing comparison

4 Biasing

4.1 Circuit

For the very initial phase of the project different biasing methods were used to bias the transistors. The first methodology used was to use the ideal voltage source to set the Direct Current (DC) Gate voltage. It is known that the real-life variations and fluctuations in real sources make this option not ideal. Another tested implementation was to bias the transistors in the circuit using current mirrors, but one of the disadvantages of this implementation is that it requires multiple current sources, that can also have variations and fluctuations. In the way it was implemented in the test bench it required multiple current sources with different current output to properly generate the bias required for the circuit, furthermore, in OTA circuits, the transconductance is really important since it impacts the gain, bandwidth and other parameters[5], as also shown in previous sections.

With the objective to not have any other power source other than the V_{DD} in the circuit, the decision of implementing a biasing circuit in order to bias all the transistors was taken.

As seen through the years and different projects and studies, one of the biggest difficulties is to have a stable reference current. One of the techniques that is used to have a stable current is to stabilize the transconductance by matching this parameter to the conductance of a resistor. By doing this, not even the variations and fluctuations in V_{DD} can affect the transconductance of the biasing circuit, which is the main reason why this biasing circuit is selected. This concept was firstly introduced by Steininger[14] and also explained by Johns and Martin[9].

The result of this process is to have a transconductance that does not depend on the power supply voltage or temperature fluctuations, making the transconductance parameter more robust against this type of changes and variations that are common in any design or component.

Analyzing the circuit on figure 7, the following expression is obtained for the transconductance of transistor *Q13*:

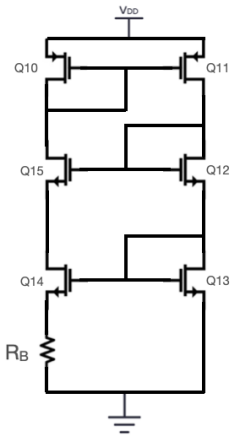


Figure 7: Stable transconductance biasing circuit [9]

$$g_{m_{13}} = \frac{2 \cdot [1 - \sqrt{\frac{(W/L)_{13}}{(W/L)_{15}}}]}{R_B}$$

As observed in the equation above, the transconductance only depends on the ratio of size ratios of the two NMOS and the resistor, thus, for the particular case where the ratio of sizes is $(W/L)_{15} = 4 \cdot (W/L)_{13}$, then the transconductance is:

$$g_{m_{13}} = \frac{1}{R_B}$$

One of the main advantages of this structure is that not only one transistor is stabilized, but all transistors in the biasing circuit. This is because the currents of all the transistors are obtained from the same biasing network, so the currents and transconductances only depend on the geometry ratios. For the NMOS:

$$g_{m_{NMOS}} = \sqrt{\frac{(W/L)_{NMOS} \cdot I_{NMOS}}{(W/L)_{13} \cdot I_{13}}} \cdot g_{m_{13}}$$

And for the PMOS:

$$g_{m_{PMOS}} = \sqrt{\frac{\mu_p}{\mu_n} \cdot \frac{(W/L)_{PMOS} \cdot I_{PMOS}}{(W/L)_{13} \cdot I_{13}}} \cdot g_{m_{13}}$$

The circuit shown in [9], the Wide-Swing Constant-Transconductance Bias Circuit, is the biasing circuit used in this thesis. The circuit is based on the circuit from figure 7, which ensures the stability of the transconductance. A start-up circuit and a cascode bias are also added.

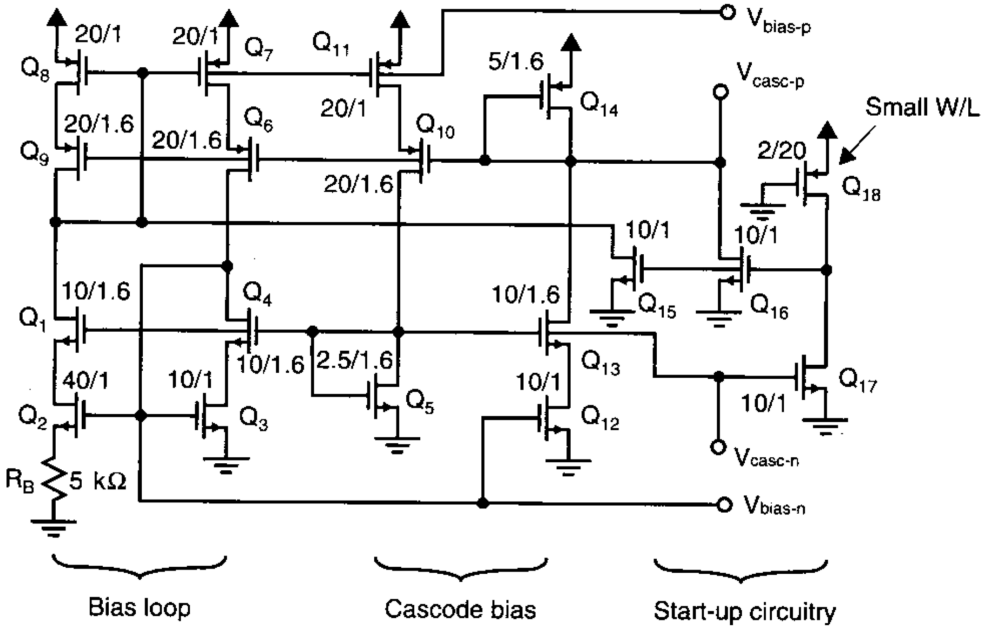


Figure 8: Wide-Swing Constant-Transconductance Bias Circuit [9]

One of the improvements of the circuit in figure 8 is the use of wide-swing current mirrors. This improvement increases the output impedance of transistors, which was not considered previously and affects negatively in the performance of the circuit in figure 7.

The cascode bias provides the gate voltage for the cascodes in the bias loop. This circuit is composed by transistors *Q10*-*Q13* and the diode-connected transistors *Q5* and *Q14*.

Another modification from the initial design is the addition of a start-up circuit. There is not a unique stable state for this circuit. One of them is when the transconductance is stable and all currents and transconductances in the circuit are stable and different from 0. But that is not the only possible stable state for this circuit. The other stable state is when all currents in the biasing circuit are zero. In that case, the biasing circuit does not generate any voltage, thus, the amplifier does not work either. This is solved with the implementation of a start-up circuit.

The start-up circuit implemented in this design only takes effect when there is no

current flowing in the biasing loop. This is because the lower transistor (*Q17*) is off, so the gates from the two transistors, *Q15* and *Q16* are turned ON and inject current into the bias loop. By injecting this current, the stable state where the circuit was is broken and the circuit switches to the other state, where the transconductance is stable. After that current is injected, transistors *Q14* and *Q9* that were initially OFF, are turned ON and transistors *Q15* and *Q16* turn OFF, so this circuit does not have any effect when the design operates in the other stable state[9].

4.2 Sizing

From previous section, it is known that the transconductance of *Q3* in the biasing circuit is fixed by the resistor, so, since the Gate-Source voltage of this transistor biases transistors *M3* and *M4* from the Conventional structure, the obtained transconductance in the sizing section of the Conventional Cascode is used to define the resistor. The initial value for the resistor was 353Ω since the transconductance obtained for transistors *M3* and *M4* was $2.83mS$.

With that resistor value, the transconductance is fixed, thus is the current. In order to get the desired overdrive voltage for the transistors *M3* and *M4* in the amplifier, through transistor *Q3* in the biasing circuit it must drive the same current, so it must be sized exactly the same as transistors *M3* and *M4* in the Conventional Structure. The size of *Q2* in the biasing circuit is fixed to be 4 times larger, to ensure a constant transconductance in *Q3*. For *Q1* and *Q4* a larger length is selected to get a larger drain-source voltage, which helps to reduce a second-order effect due to body effect and to have slightly more margin between the triode and active region [9].

For the sizing of transistors *Q9* and *Q7* in the biasing circuit, since they provide the gate-source voltage of transistor *M0* in the amplifier, they must drive the same current, so the same size is applied to both of these transistors. For *Q8* and *Q6*, the same criteria as previously applied for transistors *Q1* and *Q4* is used.

This given design was functional, nonetheless, the current consumption was large because it was using a ratio 1:1 with the circuit, so to reduce the current consumption, the size of the transistors was decreased by 4, to the ratio now 1:4, reducing the current consumption of the biasing circuit. But since the size of the transistors has changed, the value of the resistor also needs to be changed to keep the proper biasing. In this case, since the sizes of all transistors are divided by 4, the value of the resistor must be multiplied by 4 to keep the ratio constant and keep the correct biasing into the circuit.

For the sizing of the Cascode Bias section in figure 8, the same sizes as the transistors in the Bias loop are used, except for the diode-connected transistors *Q5* and *Q14*, which are set to be 4 times smaller.

For the sizing of the Start-up circuitry, the same size as *Q3* is used for all transistors except for *Q18*. For that transistor, a smaller size ratio is selected, as specified in figure 8.

The schematic of the biasing circuit is depicted in figure 9.

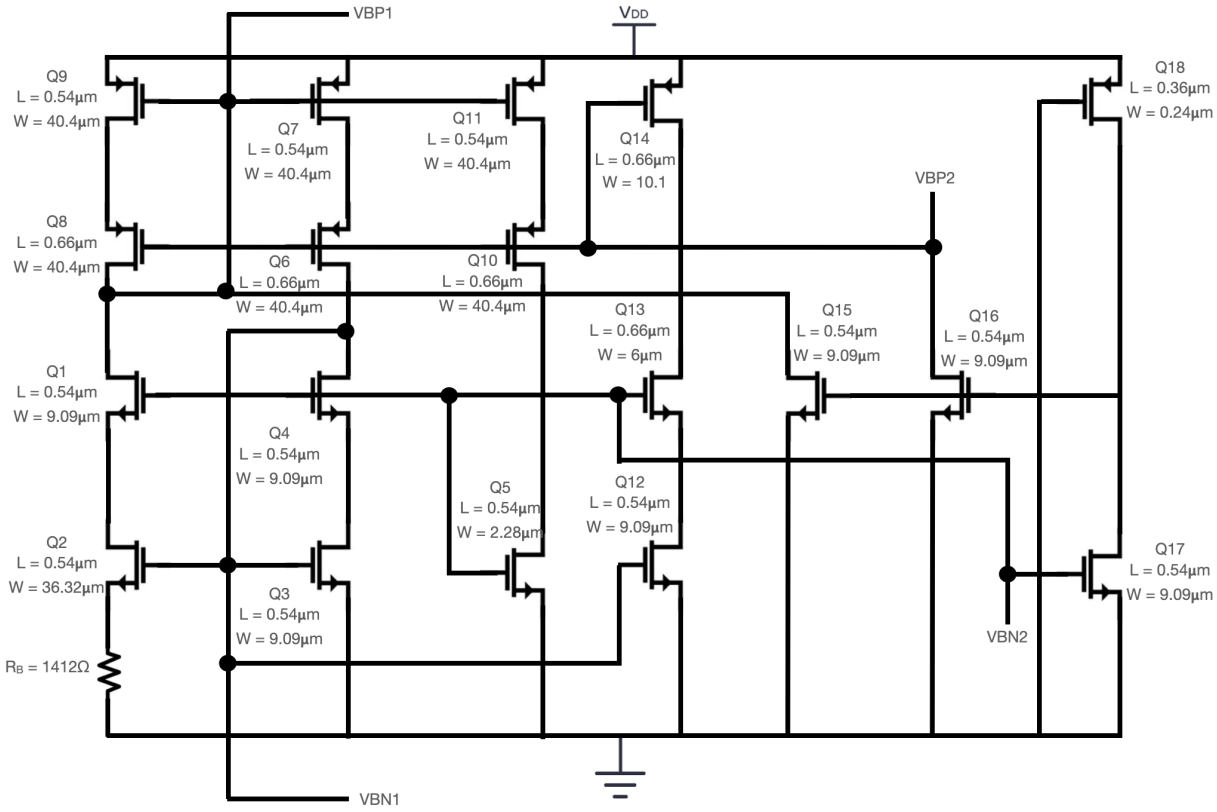


Figure 9: Biasing circuit schematic

5 Pre Layout Simulations

For the Pre Layout Simulations, both designs, the Conventional and the Recycling Folded Cascodes are checked with the same conditions and same testbench and are compared to see which one is better.

In the following subsections the circuit, the different constraints and conditions applied for the simulations are presented.

The given results correspond to the final values in the simulations. For the initial sizing, the aim was to obtain a GBW of 150MHz with a capacitance of 2.25pF for the Recycling Structure. The sizing of the Conventional structure was done with a GBW of 100MHz and the same capacitive load, nonetheless, during the first iteration of simulations, the following values were obtained:

Parameter	Conventional Structure	Recycling Structure
Bias Current (μA)	457.4	457.2
DC Gain (dB)	48.46	57.91
Bandwidth (kHz)	281.3	207.4
GBW (MHz)	74.82	154.1
Phase Margin (^{o})	80.88	57.33

Table 4: First iteration of simulations

It is observed that the GBW was reasonable, slightly above to the desired GBW for the Recycling structure (150MHz). The GBW for the Conventional structure was not reached (below 100MHz) because the value of K_P was an approximation and not the exact value. The value used for the sizing was slightly higher than the actual value, since the overdrive voltages were checked and all of them were close to 150mV. Nonetheless, the main issue observed is the Phase Margin obtained for the Recycling Structure, which is below 60°, threshold selected to define the stability.

Two approaches were attempted to increase the phase margin: change the sizing of the non-dominant pole to move it further or get the dominant pole closer to 0. By adjusting the non-dominant pole, a stable solution was achieved, nonetheless, the overdrive voltage was lower than the initial 150mV, so the other approach was preferred. The initial sizing was maintained, but 500fF were added for a total of 2.75pF, to make the Recycling structure stable. This limits the GBW and the 150MHz target is not achievable with the increase in the capacitance, but makes the design stable.

5.1 Open Loop Gain

To compute the Open Loop Gain, the circuit depicted in figure 10 is used. The open-loop frequency response is checked only with a capacitive load.

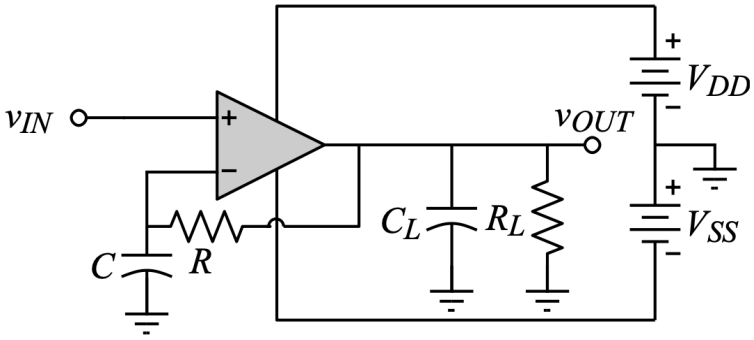


Figure 10: Circuit for Open-Loop Frequency Response[2]

The values of C and R have been set high enough thus they do not have any effect on the simulations, just to obtain a more accurate result of the Alternating Current (AC) simulation. In this case, the selected value for the capacitor was $10mF$ and for the feedback resistor was $1G\Omega$.

The simulation is set to be an AC simulation between $1Hz$ to $100MHz$.

The schematic with the values of the components is despicted in figure 11 and the results for this simulation in figure 12, where the gain in decibels is depicted for both amplifiers.

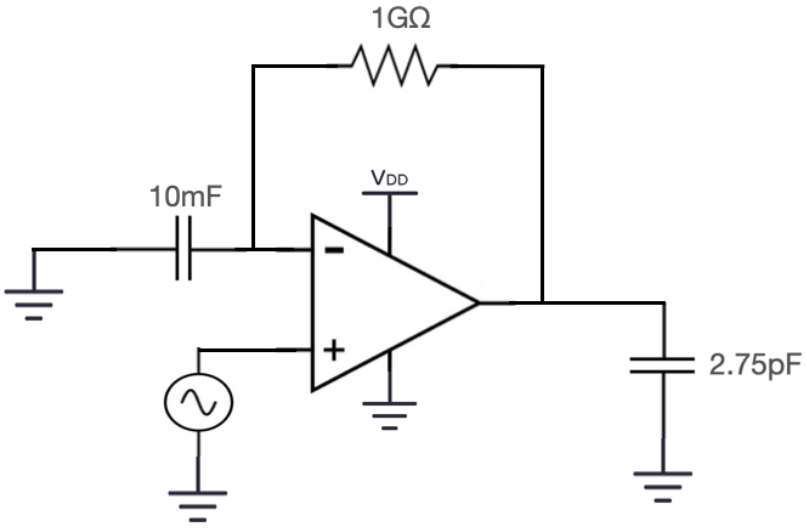


Figure 11: AC Response simulation circuit

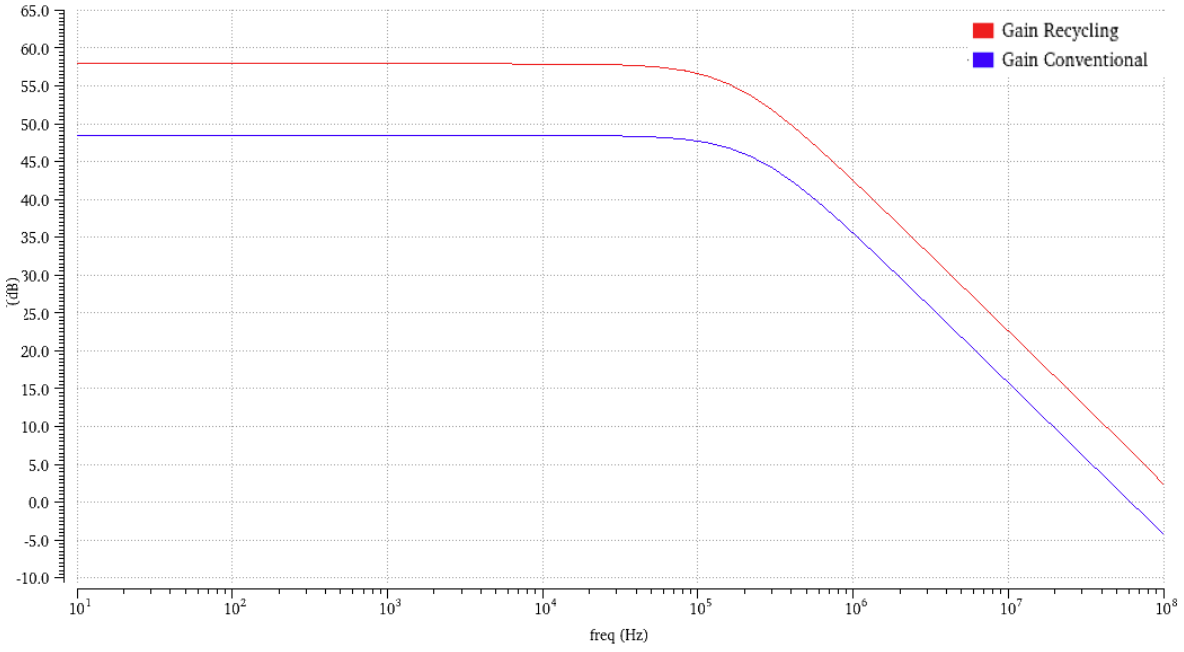


Figure 12: AC Response simulation results

	Open Loop Gain
Conventional Structure	48.46dB
Current Recycling Structure	57.91dB

Table 5: Open Loop Gain values for the pre layout simulations

Looking at the simulation results, it is concluded that the gain of the Recycling Structure is 57.91dB and the Conventional Structure is measured at 48.46dB. As mentioned in previous sections and in the specifications of the thesis, for a fair comparison, both amplifiers have a similar current consumption, as shown in table 17, so in this parameter it can be clearly stated that the Recycling outperforms the Conventional Structure. The increase in the gain is, as mentioned also in the comparison between both structures, higher than 6dB due to the increase in the transconductance. As mentioned, because of the increase in the output resistance, a slightly higher gain increase is expected.

5.2 Bandwidth

To measure the Bandwidth of the amplifiers, the same circuit for the Open Loop Gain is used and the same response is analyzed. To get the Bandwidth of the amplifiers the point where the $-3dB$ point is reached is used.

For a better precision in the bandwidth calculation, the calculator is used to get the value using the bandwidth function. The expression used is the following:

bandwidth((VF("output") / VF("input")) 3 "low")

As seen in table 6, the bandwidth of the Conventional structure is higher than the Current Recycling.

	Bandwidth
Conventional Structure	233.9 kHz
Current Recycling Structure	173.2 kHz

Table 6: Bandwidth values for the pre layout simulations

Nonetheless, the parameter that actually shows the performance of the amplifier is the GBW, which is calculated from the gain and the bandwidth obtained. In this case, for a better accuracy, the calculator is used with the following expression:

unityGainFreq(VF("output") / VF("input"))

The results obtained are shown in table 7. It is observed that with the same current consumption, the current recycling is clearly outperforming the conventional structure by having more than the double of GBW. Nonetheless, due to the increase of the capacitance load, the specification of reaching 150MHz is not satisfied, but the stability of the design is prioritized and the obtained GBW reduction is tolerable.

	Gain-Bandwidth Product (GBW)
Conventional Structure	63.66 MHz
Current Recycling Structure	133.2 MHz

Table 7: GBW values for the pre layout simulations

5.3 Phase Margin

From the initial simulations, it was proved that the added transistors in the Recycling Folded Cascode degrade the Phase Margin, making the system unstable. By adding a capacitor, the system was brought to stability, but it is an important factor to consider when switching to the Recycling Structure. Also, the stability of this structure has a dependency on the load, which does not happen in the Conventional Structure, because the Phase Margin is large enough that the changes in the load do not highly affect the stability.

For this measurement, as previously done, for a more precise value, the calculator is used to obtain the values. The expression is:

- Phase Margin: *phaseMargin((VF("output") / VF("input")))*

The Complete AC response where the Phase Margin is obtained for both amplifiers is depicted in figure 13.

The Phase Margin obtained with the expression is shown in table 8 and both amplifiers are stable, nonetheless, there is a clear degradation in the Phase Margin, which must be considered if using the Recycling structure.

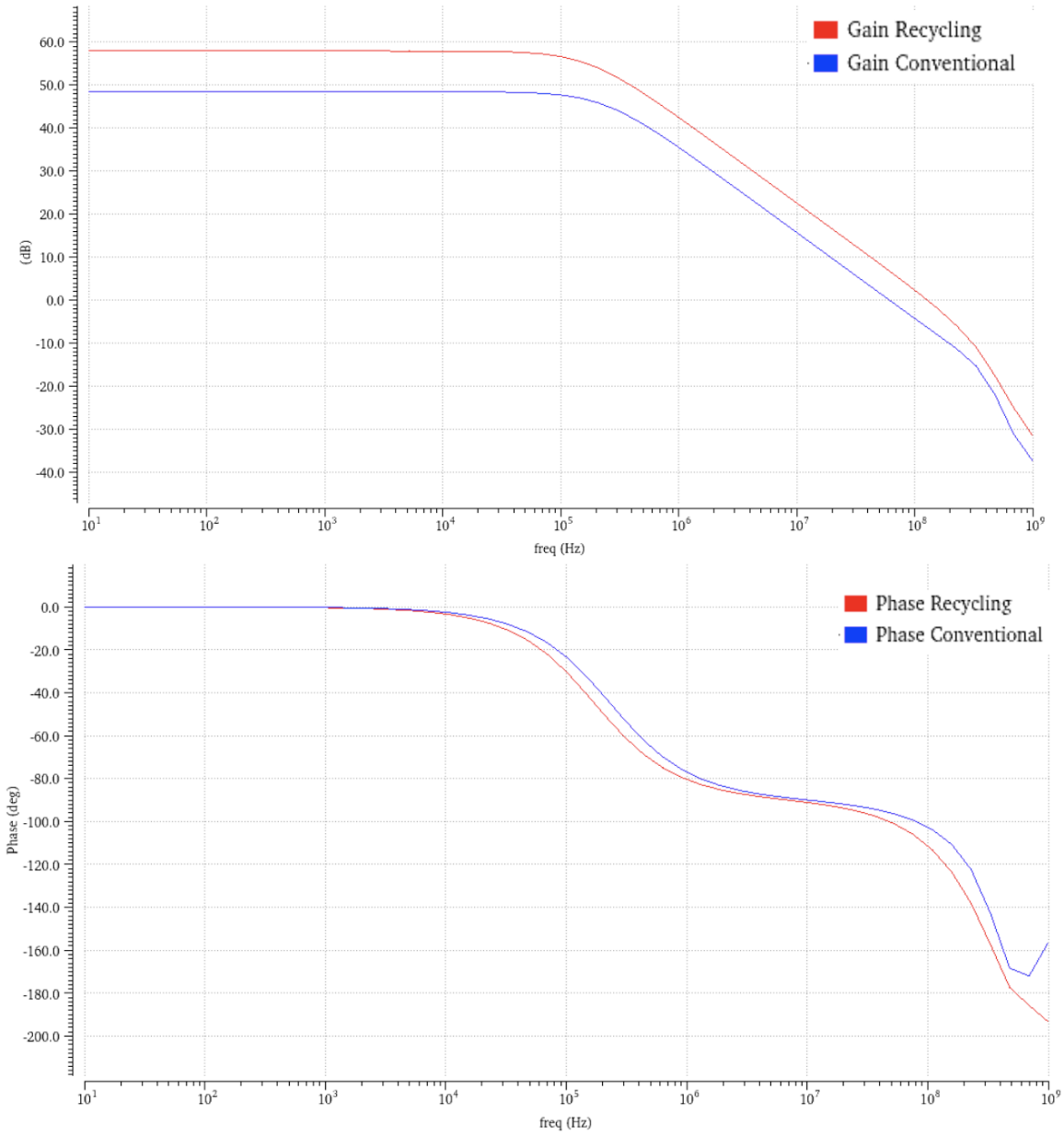


Figure 13: Complete AC Response for Margin calculations

	Phase Margin
Conventional Structure	82.49°
Current Recycling Structure	62.77°

Table 8: Phase Margin for the pre layout simulations

5.4 DC Offset Voltage

The DC offset voltage is obtained with the circuit in figure 14

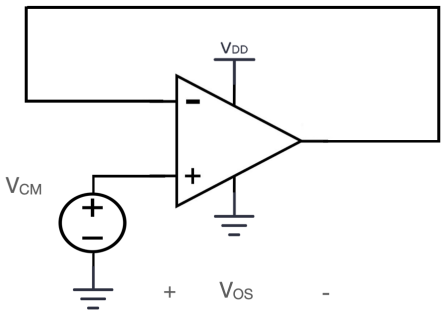


Figure 14: Schematic for measuring the DC Offset Voltage

This parameter is important to get a more accurate Common Mode Rejection Ratio (CMRR) result and to check which is the level of DC Offset that the amplifier has for a better characterization of the performance.

To get this value, a DC simulation is done and the voltage between V_{CM} and the output is measured to obtain the DC Offset voltage.

Refer to table 9 for the obtained DC Offset voltage for both amplifiers. The Recycling Structure has a smaller offset magnitude than the Conventional.

	DC Offset Voltage
Conventional Structure	-942.3μV
Current Recycling Structure	-319μV

Table 9: DC Offset Voltage for the pre layout simulations

5.5 Common Mode Rejection Ratio (CMRR)

Because both designed amplifiers are differential, the CMRR is an important characteristic to check the ability to reject common-mode signals for both amplifiers. To do so, the circuit in figure 15 is used. The offset voltage obtained in previous subsection is used to compensate this voltage and get a more accurate CMRR value.

By definition, the CMRR is calculated as:

$$CMRR = \frac{A_v}{A_{CM}} = \frac{V_{out}}{V_{CM}} \tag{24}$$

Where V_{out} is the output of the differential amplifier, that is obtained from the open-loop gain simulation and V_{CM} is the output of the Common Mode that is obtained from the circuit in figure 15.

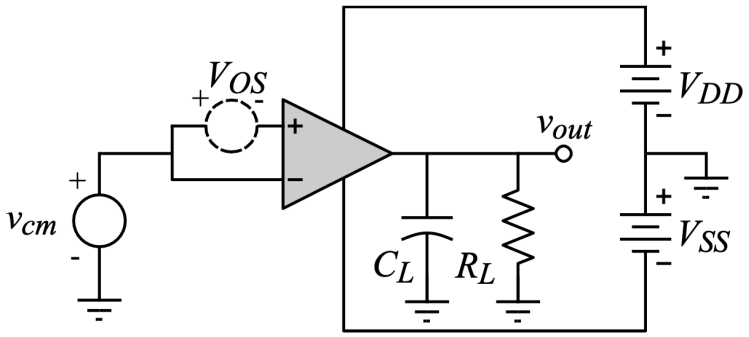


Figure 15: Circuit for Common-Mode Rejection Ratio [2]

The CMRR is checked at $10Hz$ and at $100MHz$.

An AC simulation is performed to obtain the AC behavior in Common Mode. The schematic used is given in figure 16. The value of the DC source in the input is changed accordingly to the DC Offset Voltage obtained in the previous section. The result is shown in figure 17, where it is observed that the CMRR of the Recycling structure is higher, except for a small range where the conventional is slightly better. That behavior is expected because the gain of the Recycling structure is higher, thus, when dividing by the Common Mode Gain, which is expected to be small, the structure with higher gain is also expected to have higher CMRR.

	CMRR @ 10Hz	CMRR @ 100MHz
Conventional Structure	63.29dB	34.53dB
Current Recycling Structure	69.72dB	37.06dB

Table 10: CMRR for the pre layout simulations

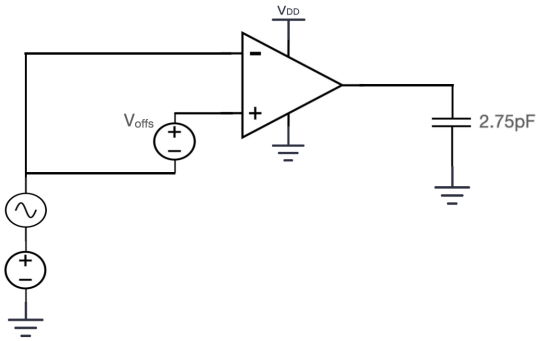


Figure 16: Schematic for CMRR simulation

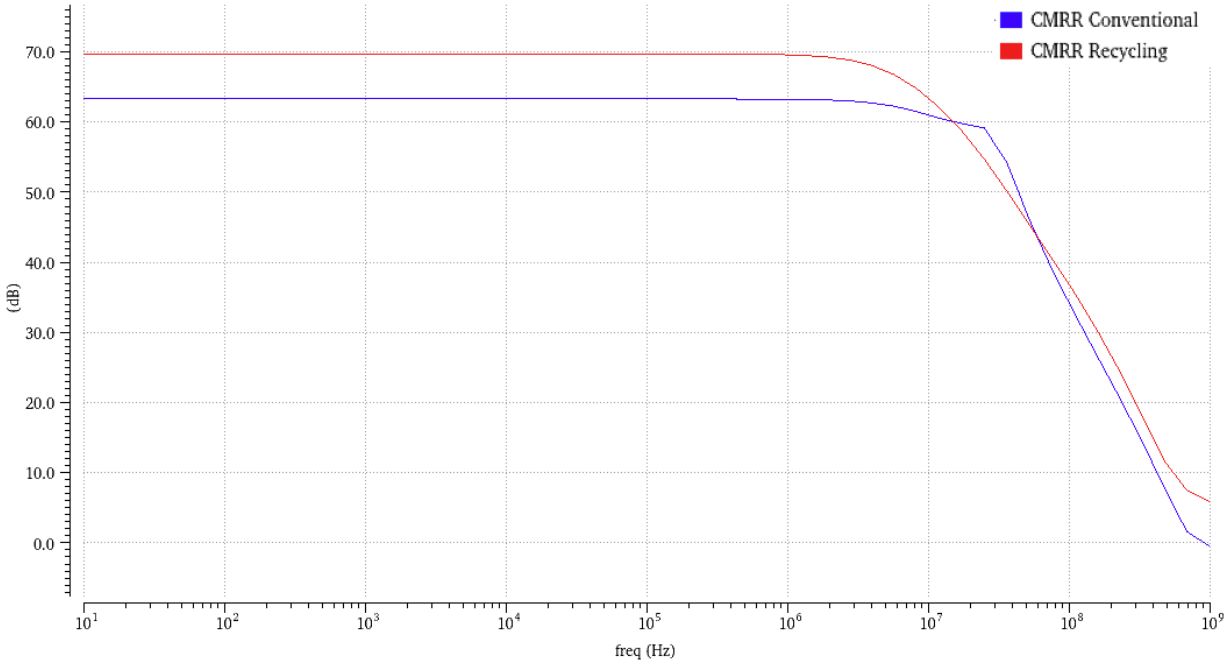


Figure 17: Result for CMRR simulation

5.6 Power Supply Rejection Ratio (PSRR)

Given that it is possible that the power supply of the amplifier has some variations, it is important to check the PSRR to check the ability of both amplifiers to suppress any variation in the power supply that can affect the output signal. To do that, the circuit used is depicted in figure 18 and the following expressions are used:

$$PSRR+ = \frac{V_{dd}}{V_{out}} \tag{25}$$

$$PSRR- = \frac{V_{ss}}{V_{out}} \tag{26}$$

Where V_{out} is the output, and V_{dd} and V_{ss} is the ripple in the inputs.

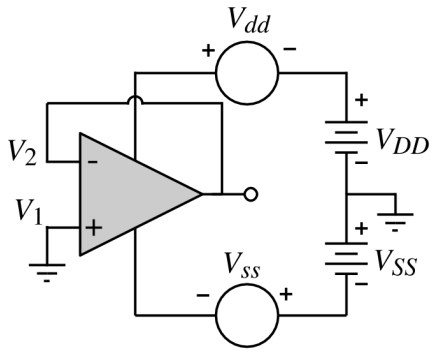


Figure 18: Circuit for Power Supply Rejection Ratio [2]

The Schematic used and the results for both amplifiers is shown in figure 19. It is observed that the PSRR is higher for the Recycling structure, which is expected. This is because the definition of the PSRR is the relation between the input gain and the supply noise gain. Even both amplifiers have a similar noise gain, since the loop gain of the Recycling structure is higher, it also leads to higher PSRR.

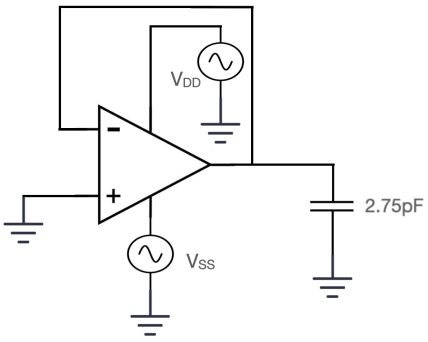


Figure 19: Schematic for Power Supply Rejection Ratio simulation

The results are shown in figures 20 and 21.

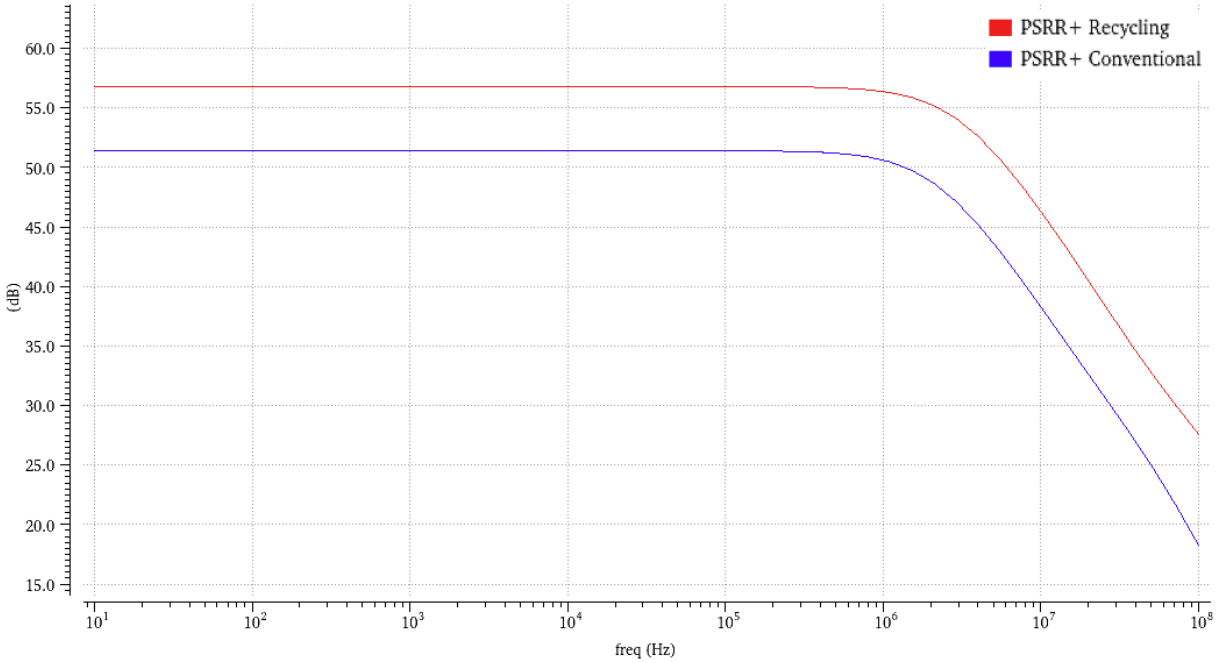


Figure 20: Result of PSRR+ simulation

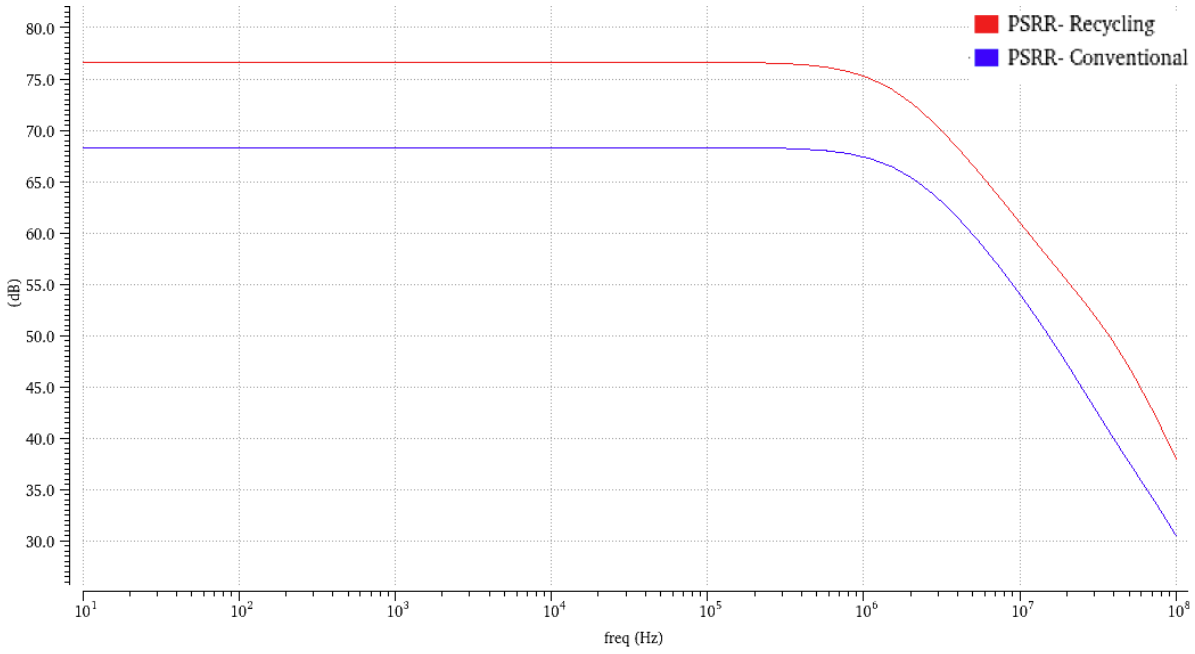


Figure 21: Result of PSRR- simulation

	PSRR+ @ 10Hz	PSRR- @ 10Hz
Conventional Structure	51.40dB	68.34dB
Current Recycling Structure	56.79dB	76.65dB

Table 11: PSRR for the pre layout simulations

5.7 Input Common Mode Range (ICMR)

The range of input that the amplifier is working in the intended region (saturation) is defined by the ICMR. To measure this parameter, the circuit from figure 22 is used.

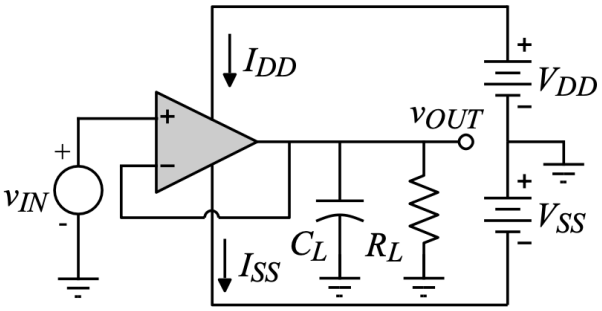


Figure 22: Circuit for Input Common Mode Range [2]

The valid range is given for the range of voltages where the output has slope 1. As a reference, the DC voltage is also plotted, because it has slope 1. The result is plotted in figure 24 and the schematic used in figure 23. A DC simulation is done with a sweep in the DC value of the input voltage. The voltage is swept from -0.5V to 2V and when the input value deviates a 2% from the input, it is considered that the amplifier is not working in the intended region anymore.

The results show that the input range where the amplifier works in the saturation region is larger in the Recycling structure compared to the Conventional.

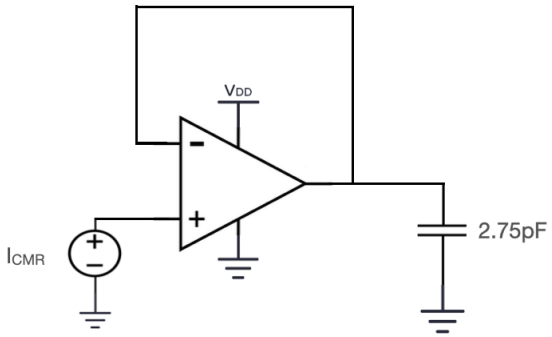


Figure 23: Schematic for Input Common Mode Range

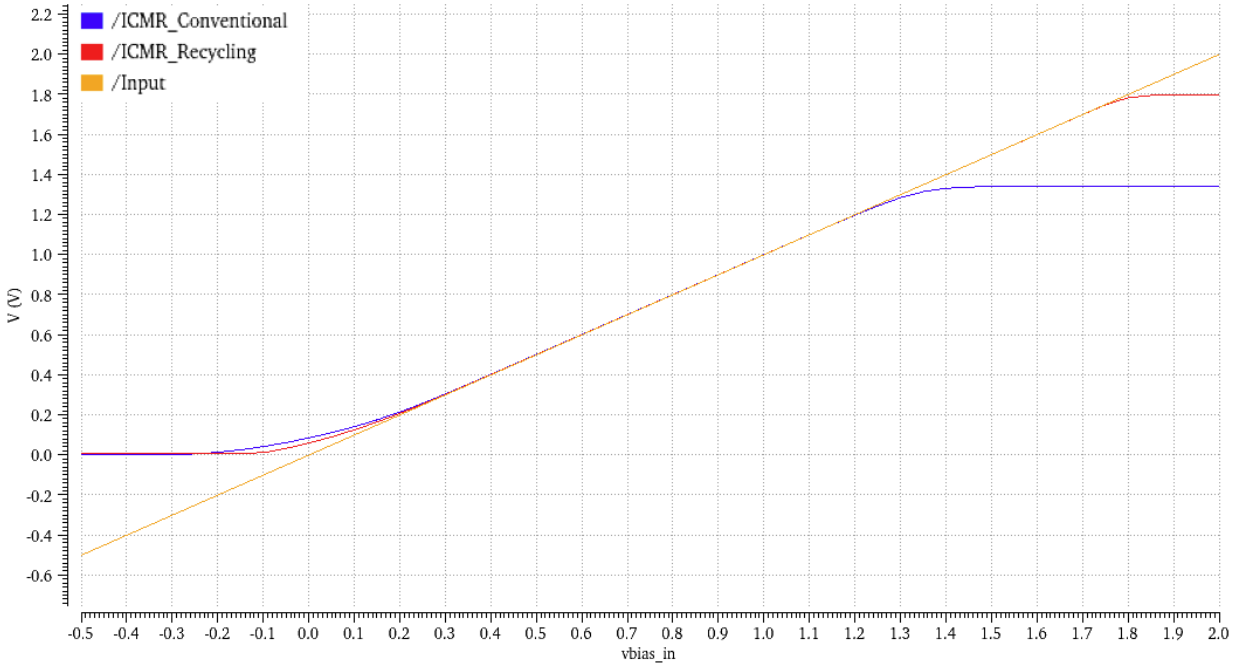


Figure 24: Results for the pre layout Input Common Mode Range

	ICMR [V]
Conventional Structure	[0.3; 1.27]
Current Recycling Structure	[0.3; 1.74]

Table 12: ICMR for the pre layout simulations

5.8 Slew Rate

For the measurement of the Slew Rate, the same circuit than for the Open-Loop simu-

lations is used, but instead of using the sinusoidal voltage source, a pulse source is used. A transient simulation is performed in order to assess the slew rate of the amplifier. The circuit used is given in figure 25.

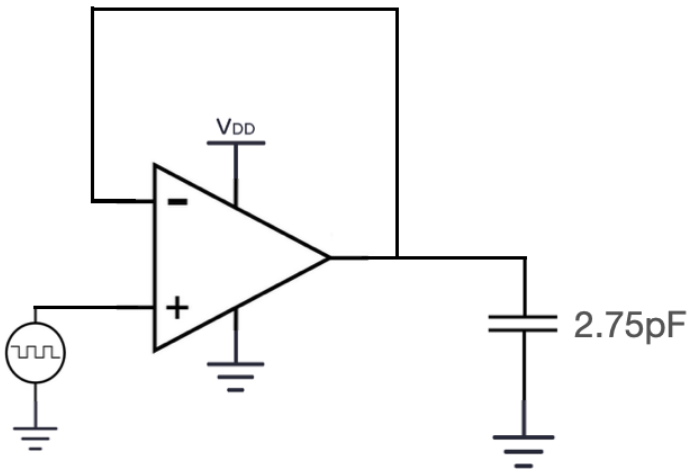


Figure 25: Circuit for transient simulations

The transient response of the amplifiers is shown in figure 26.

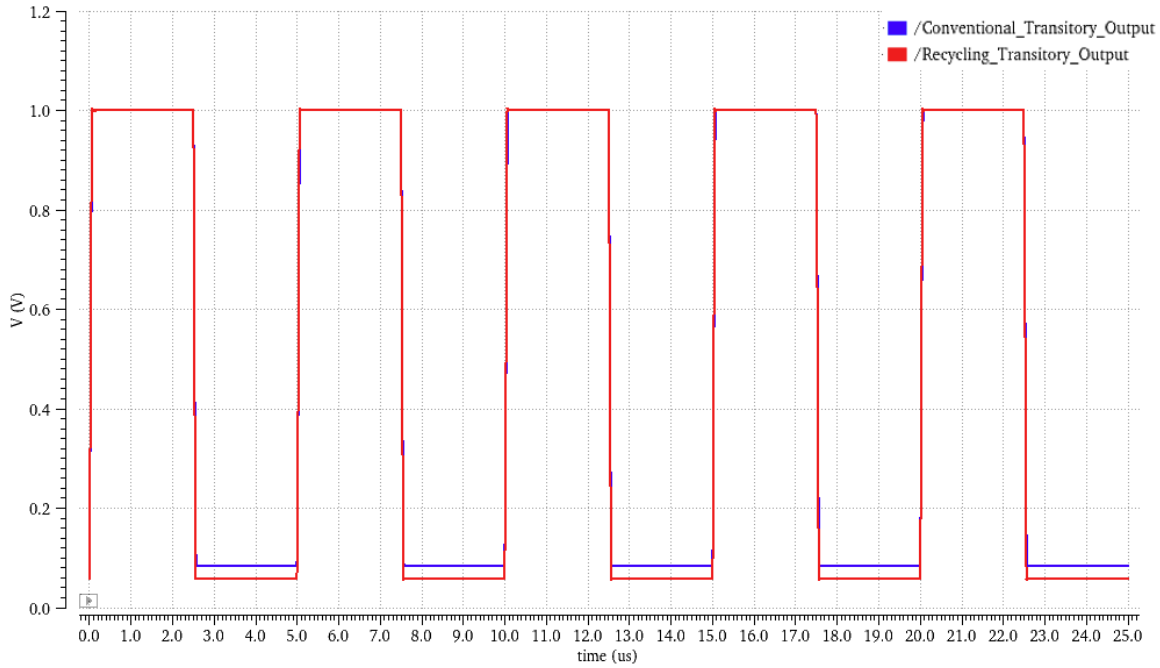


Figure 26: Transient response

Besides the visual check of the Slew Rate measurement, a calculator expression is used to check the value of the Slew Rate for a more precise calculation. From the visual check shown in figure 27, the Slew Rate of the Recycling is slightly higher than the Conventional, but not as much as the theory expects.

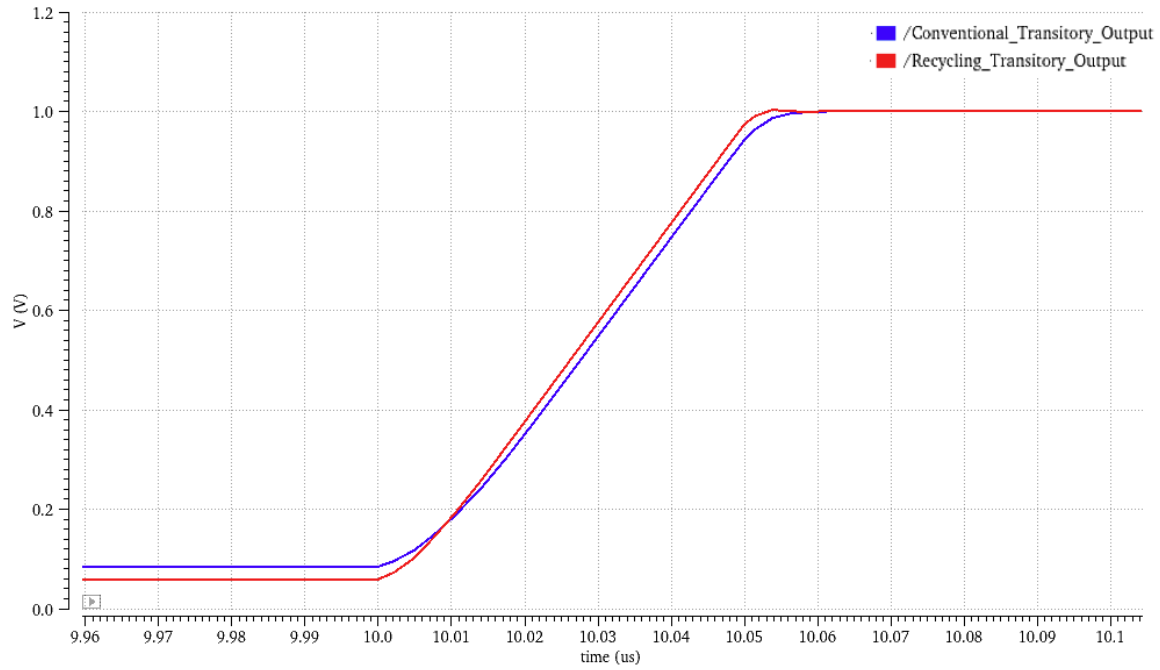


Figure 27: Slew Rate visual check

The calculator expression used are and the output results is given in table 13:

Positive Slew Rate: *slewRate(VT("output") 0 nil 1 nil 10 90 nil "time")*

Negative Slew Rate: *slewRate(VT("output") 1 nil 0 nil 10 90 nil "time")*

	Slew Rate +	Slew Rate -
Conventional Structure	17.69 V/μ s	-18.4 V/μ s
Current Recycling Structure	19.16 V/μ s	-18.95 V/μ s

Table 13: Slew Rate for the pre layout simulations

5.9 Settling time

For both amplifiers, two settling times are computed. The first one is the $10\% - 90\%$ and the other one is the measure of the overshoot. The same circuit and simulation as the

Slew Rate simulation is used, but other part of the transient is checked and measured. For both measurements, markers are used at 0.1V and 0.9V, given that the voltage pulse used is 1V. For the overshoot, an amplified view of the transition is taken to measure it.

Regarding the overshoot, it is observed that the Conventional Structure does not have any overshoot, it always keeps below the expected voltage, whereas the Recycling structure has a slight overshoot of 0.12% as shown in figure 29. Since the overshoot is not high enough to need time to stabilize around 1%, the overshoot magnitude is used to compare against the post layout.

The results of the settling time and the overshoot are:

	Settling time (90% Final Value)	Overshoot
Conventional Structure	44.67ns	0%
Current Recycling Structure	41.95ns	0.12%

Table 14: Settling time and overshoot for the pre layout simulations

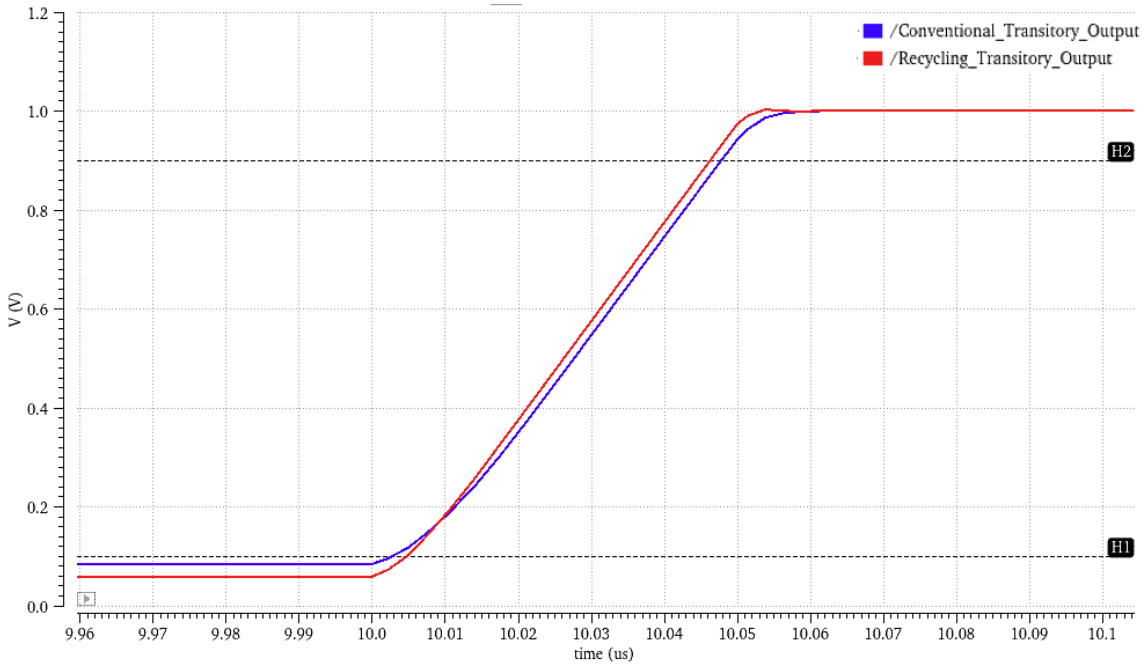


Figure 28: Settling time markers

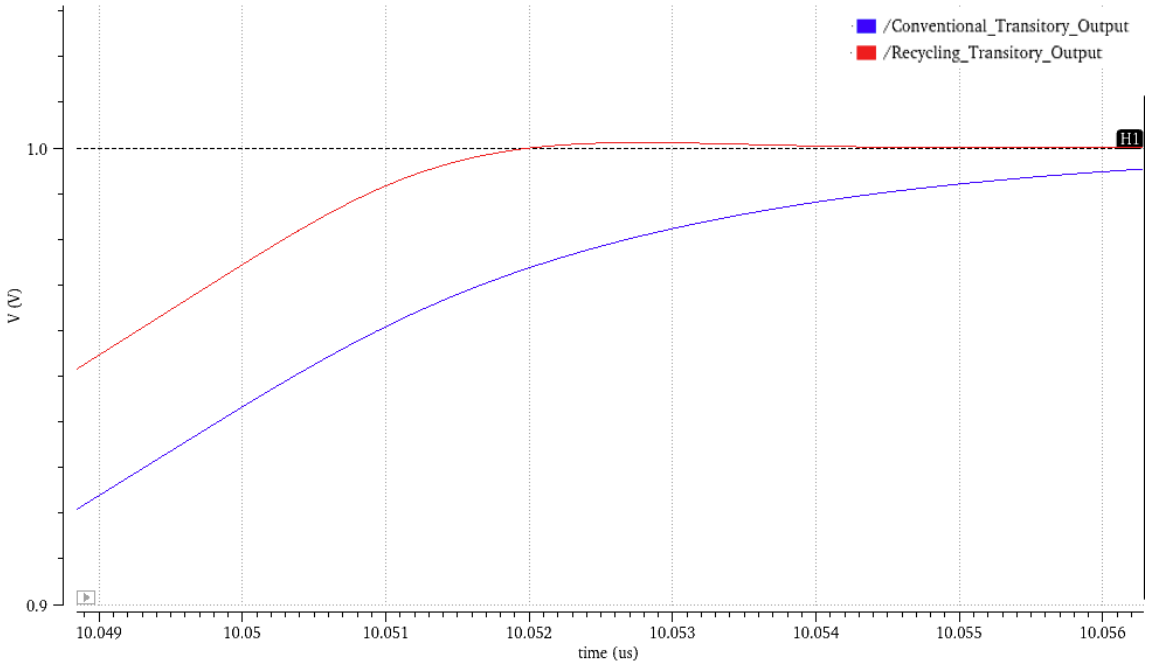


Figure 29: Close look at the overshoot

5.10 Linearity

For the linearity measurement the second and third Harmonic Distortion (HD) (HD2 and HD3) are measured. The procedure followed is to calculate the Discrete Fourier Transform (DFT) and from that, calculate the magnitude of the DFT in decibels.

Since the frequency of the source is known, in this case set to 1MHz as shown in figure 30 the second and third harmonics are 2MHz and 3MHz respectively. The magnitude of these harmonics is subtracted to the fundamental and the HD2 and HD3 are calculated.

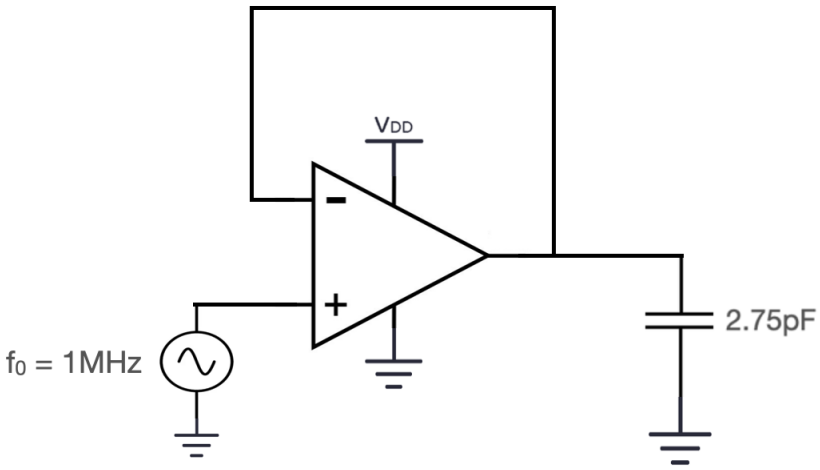


Figure 30: Schematic for linearity simulations

The obtained values are:

	HD2	HD3
Conventional Structure	56.07dB	61.05dB
Current Recycling Structure	55.04dB	59.85dB

Table 15: HD2 and HD3 results for the pre layout simulations

The figures showing the DFT for both systems are figure 31 and 32 for the Conventional and the Recycling structure respectively. Conventional structure has a slight advantage in terms of linearity against the Recycling structure.

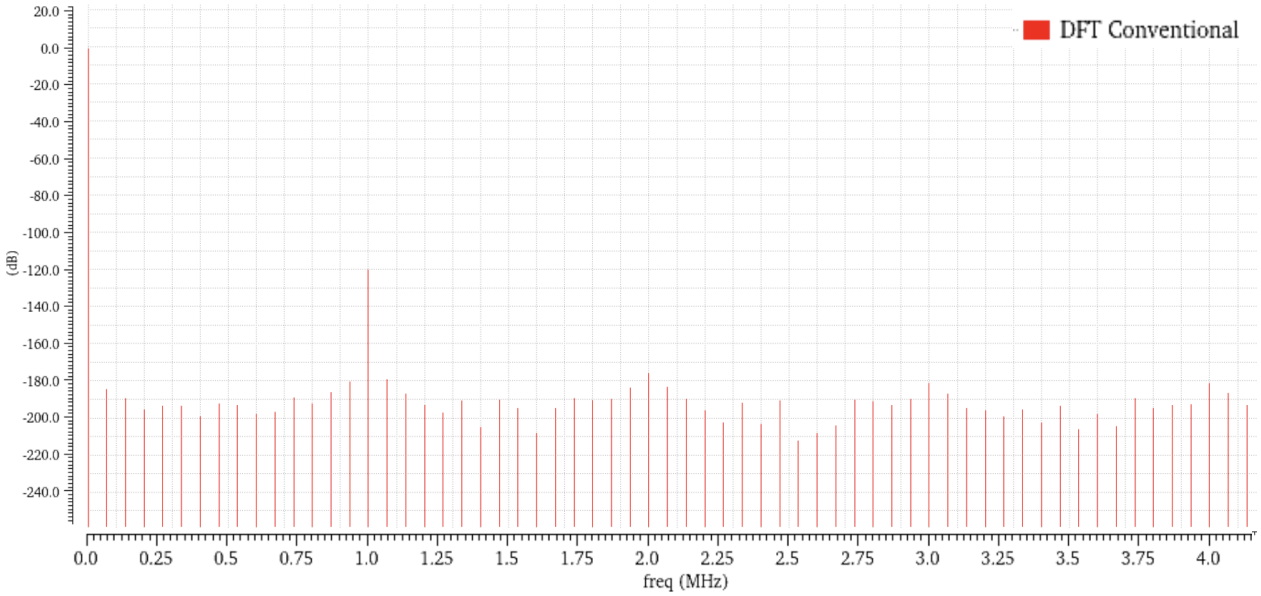


Figure 31: DFT result for Conventional Structure

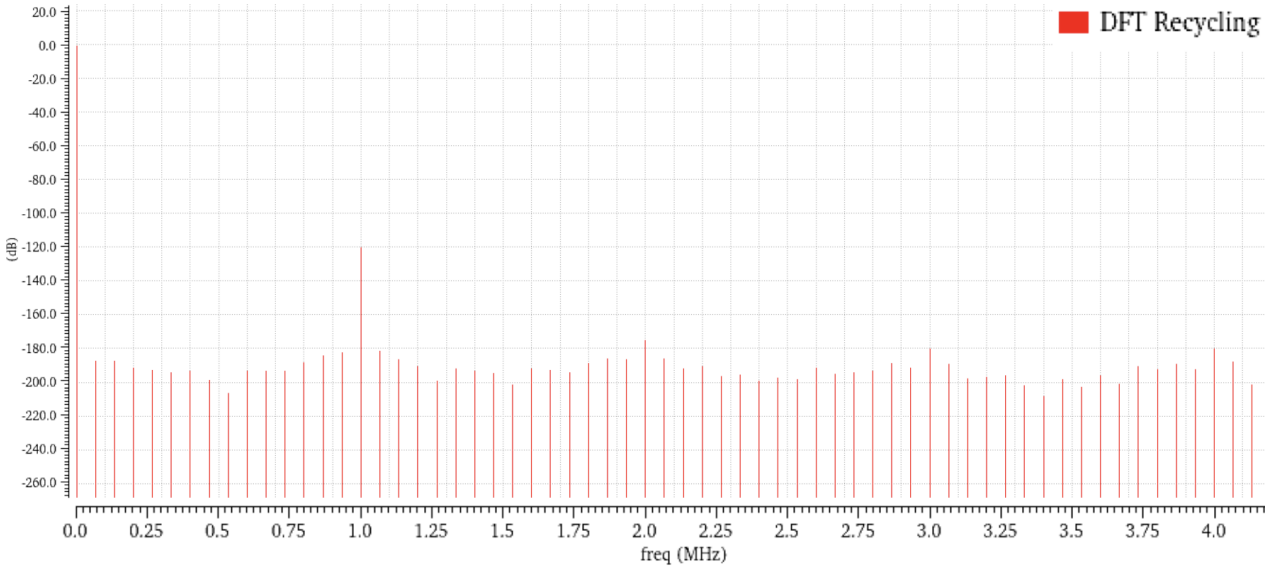


Figure 32: DFT result for Recycling Structure

5.11 Noise

For the noise measurements of both amplifiers, the schematic depicted in figure 30 is used. Since Virtuoso has the noise performance simulation, it is used to characterize the noise of both amplifiers. Given that the simulation outputs the value of the output referred noise, but not for the input, the calculator is used for that measurement.

- Input referred noise: *sqrt(integ((getData("in" ?result "noise")**2) 1 100000000))*

Two different simulations have been done, given that in the simulation the output and input nodes must be specified.

The results for the Root-Mean-Square (RMS) input referred noise between 1Hz and 100MHz are shown in table 16 and the input referred noise curve in figure 33. The input referred noise shows that the Conventional structure is a noisier structure. That is also appreciated in the noise curves, where the values for the Conventional structure are larger.

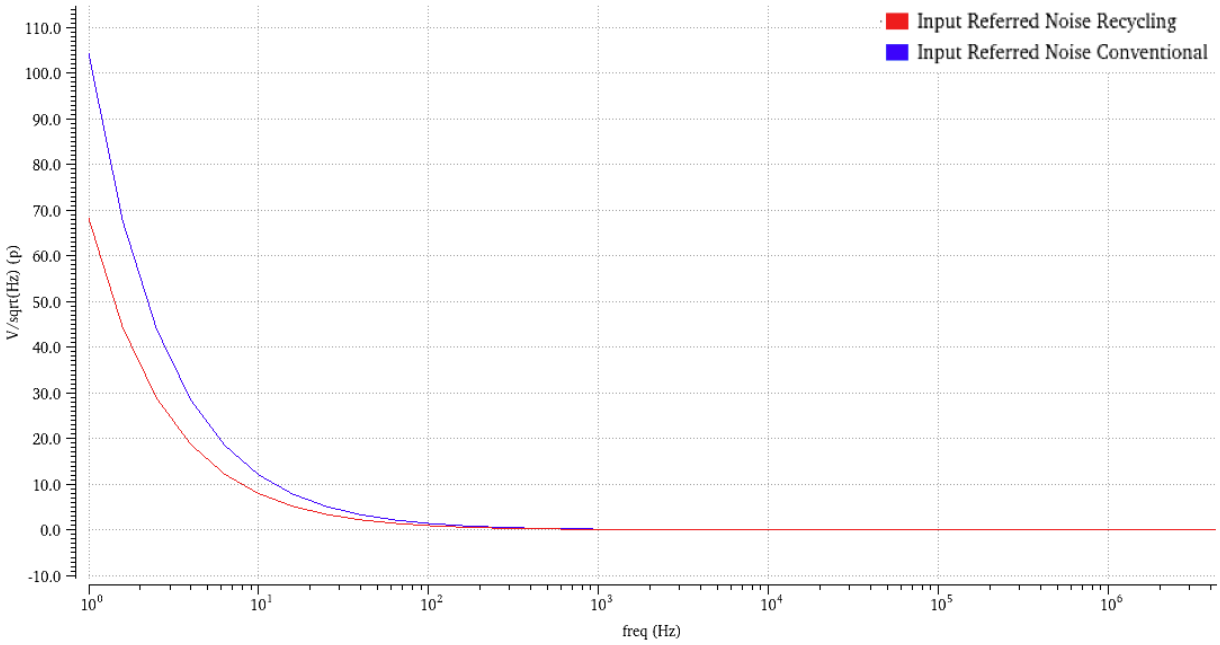


Figure 33: Output Referred noise curve for Conventional Structure

	Input Referred Noise (Vrms)
Conventional Structure	107μV
Current Recycling Structure	87.45μV

Table 16: Noise for the pre layout simulations

5.12 Comparison and simulation conclusion

Table 17 summarizes the simulations and checks performed in this section. The values that are better for each parameter are highlighted. The Bias Current does not have a subsection, but it was checked with the operating point and only the amplifier current is considered.

The Recycling Folded Cascode with similar current consumption is clearly outperforming the Conventional Folded Cascode in terms of DC Gain, where the approximated 8-9dB increased is shown. Given that increase in the gain, the GBW it is slightly over the double despite a decrease in the bandwidth.

The Slew Rate performance of the Recycling Structure is better than the Conventional, nonetheless, the theoretical improvement was that the Recycling should be three times larger than the Conventional, however, the actual improvement obtained from simulation shows a smaller improvement.

Another parameter that has improved from the conventional to the recycling is the PSRR. The increase in the PSRR is expected, given that the gain in the Recycling structure is higher. Also because of the increase in the GBW the improvement in the PSRR extends to higher frequencies.

The input referred noise is also lower for the Recycling structure in comparison with the Conventional structure.

CMRR is higher in the Recycling Structure and the input where the amplifier is working in saturation has also slightly increased, covering a higher input range.

Nonetheless, besides the expected worse Phase Margin, which clearly shows up with a decrease of roughly 20°, which must be considered when designing given that could lead to really low Phase Margin and bring the system to a non-stable behavior, as in the initial simulations.

Furthermore, another parameter where the Conventional Structure shows a slight performance improvement against the Recycling is the linearity, where the distortion of the harmonics is higher for the Conventional.

Despite the above mentioned decrease in performance, the improvements are more, so after the pre layout simulations, it is concluded that the Recycling structure is a structure that is worth switching to, since the performance improvement justifies the parameters performance decrease and the modifications to switch from the conventional to the recycling are not extremely difficult to perform, which have a similar power consumption and a slight increase in the area, given the two extra added transistors and the split transistors. Nonetheless, the stability of the design must be monitored to avoid non-stable behavior.

Parameter	Conventional Structure	Recycling Structure
Bias Current (μA)	457.4	457.2
DC Gain (dB)	48.46	**57.91**
Bandwidth (kHz)	**233.9**	173.2
GBW (MHz)	63.66	**133.2**
Phase Margin (^{o})	**82.49**	62.77
Slew Rate + ($V/\mu s$)	17.69	**19.16**
Slew Rate - ($V/\mu s$)	-18.4	**-18.95**
Offset Voltage (μV)	-942.3	**-319**
PSRR+ @ 10Hz (dB)	51.40	**56.79**
PSRR- @ 10Hz (dB)	68.34	**76.65**
CMRR @ 10Hz (dB)	63.29	**69.72**
CMRR @100MHz (dB)	34.53	**37.06**
ICMR [V]	[0.3; 1.27]	[**0.3; 1.74**]
Settling time (90% Final Value) (ns)	44.67	**41.95**
Overshoot	**0%**	0.12%
Input referred noise [μVrms] 1Hz - 100MHz	107	**87.45**
HD2 (dB)	**56.07**	55.04
HD3 (dB)	**61.05**	59.85

Table 17: Characterization comparison between Conventional and Recycling structures

6 Layout

Given that the amplifier has a differential pair and current mirrors, the matching between transistors is an important factor to be considered during the layout of the device. The criteria for the layout focuses on this idea of matching transistors in the layout.

This section specifies the criteria followed in order to proceed with the layout and the resulting layout and the righteousness of the decisions taken for each block of the layout.

6.1 Layout criteria

The main criteria followed for the layout are:

1. Although chip area is not an issue, whenever is possible, try to compact as much as possible the layout.

2. No restriction in the usage of available metal layers, but avoid higher layers to keep the P-type and N-Type (PN) junctions hidden from exposition to light to avoid PN junctions variations. The reason of that is because one of the potential usages of this amplifier is in combination with optical sensors and this amplifier must be covered with higher metals to prevent to be exposed to light. The top metal is not used in the layout.

3. Larger transistors are divided into smaller ones to avoid excessively large transistors layout.

4. For improved matching, the maximum simmetry is kept among transistors, by keeping rectangular shapes and the same orientation: the transconductance of Metal-Oxide-Semiconductor (MOS) transistors depend on carrier mobilities, which show orientation-dependent sensitivities. Having MOS transitors oriented in different directions leads to different intrinsic transconductances for each of those transistors making the matching difficult due to mismatches in the current[7].

5. Another aspect to be considered is the gradient-induced mismatches. Mismatch caused by gradient is dependant on the separation between the effective centers, or centroids, of matched devices. Mismatch due to gradient depends on the magnitude of the gradient and also the distance between the centroids of the matched devices. To minimize this effect, the **common-centroid layout** is applied. Applying common-centroid layout cancels the effects of long-range variations as long as these are linear functions of distance, or even if the variations have non-linearities, a linear approximation is considered over short distances[10].

6. As general norm, transistors should be placed thus the current follows the same direction[7].

7. An exception of previous criteria is when transistors can be interdigitated, when transistors can share a source or drain diffusion. Interdigitation ensures exact alignment of the centroids. By applying interdigitation, previous criteria for layout is not fulfilled, since in a pattern of 2 interdigitated transistors in the form ${}_{D}A_{S}B_{D}B_{S}A_{D}$, as shown in the example in figure 34, it is observed that the current through the transistors of the same segment have the current flowing in opposite directions, nonetheless, if in an interdigitated block a transistor is composed by the same number of left-oriented and right-oriented segments, the orientation effects that lead to mismatch are canceled. The before-mentioned orientation effects can take place because of diagonal shifts when ion implantation is performed at an angle. Tilted implants do not have a big impact in matched transistors operating in the linear area, nonetheless, in saturated devices it can lead to some transconductance mismatch.[7]

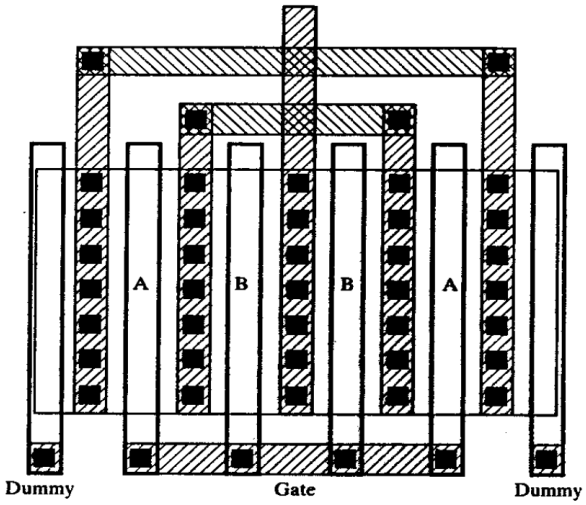


Figure 34: Interdigitated MOS transistors example[7]

8. Matched transistors are placed in close proximity.

9. Only paired transistors are designed to match.

6.2 Amplifier

For the layout of the transistor groups listed below, a matching technique is used:

1. Input transistors

Input transistors are split in 4 transistors each, for a total of 16 input transistors. For the common-centroid and interdigitation pattern to ensure a good matching in the transistors the following structure is followed:

$$_{S}W_{D}W_{S}X_{D}X_{S}Z_{D}Z_{S}Y_{D}Y_{S}Y_{D}Y_{S}Z_{D}Z_{S}X_{D}X_{S}W_{D}W_{S}$$

The legend of the transistors is:

- Transistor M1A: W
- Transistor M1B: X
- Transistor M2A: Y
- Transistor M2B: Z

Refer to figure 35 for the implemented layout.

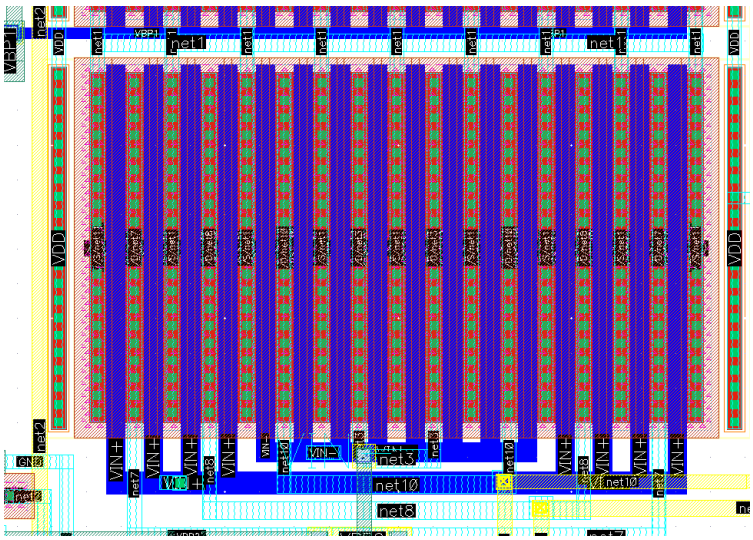


Figure 35: Layout of input transistors

2. Current mirror in M3 and M4

For this structure, the transistors are split. All the transistors have the same width and length, but since one of the branches must be 3 times larger, one of the branches

has the triple amount of transistors than the other. In this case one branch has 4 transistors, whereas the other has 12.

For a more compact solution a 32x1 structure is implemented. A 2-D Common Centroid Layout was also considered [10] but the space that was required between rows to do the connections was more space consuming than making a 32x1. Refer to figure 36 for a closer look at the layout.

$$_SX_DX_SZ_DZ_SW_DW_SW_DW_SW_DW_SY_DY_SY_DY_SY_DY_SY_DY_SY_DY_SY_DY_SW_DW_SW_DW_SW_DW_SZ_DZ_SX_DX_S$$

The legend of the transistors is:

- Transistor M3A: W
- Transistor M3B: X
- Transistor M4A: Y
- Transistor M4B: Z

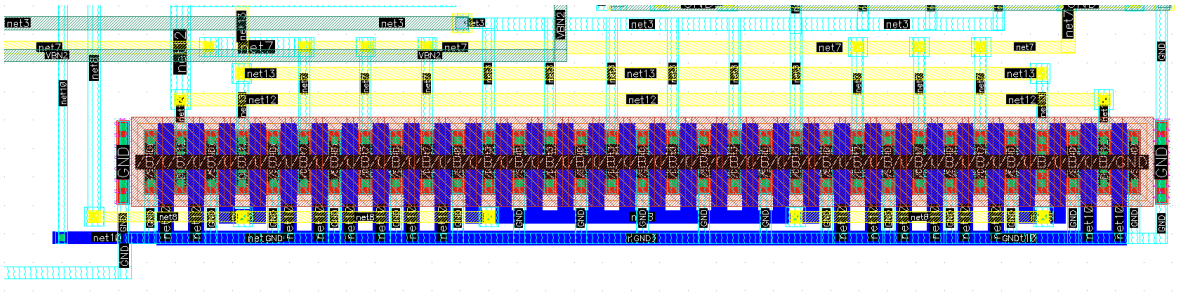


Figure 36: Layout of current mirror

3. Transistors M11 and M12

These transistors only are placed in a common-centroid structure, given that inter-digitation is not possible. Transistors are placed as follows:

$$YZZY$$

- Transistor M11: Y
- Transistor M12: Z

The layout is depicted in figure 37.

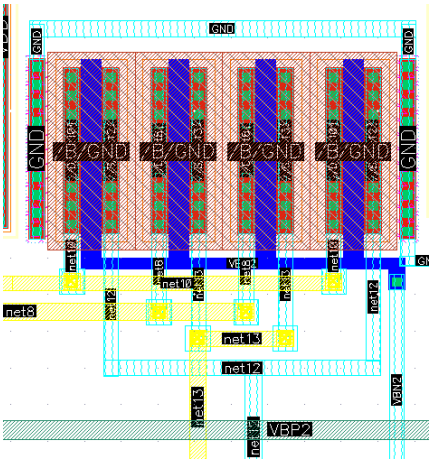


Figure 37: Layout of transistors M11 and M12

4. Transistors M5 and M6

Same case and structure as transistors M11 and M12, but in this case split in 8 transistors in the structure $YYZZZZYY$. The size of transistors is the same, but in this case there is double of each. Layout is given in figure 38.

- Transistor M5: Y
- Transistor M6: Z

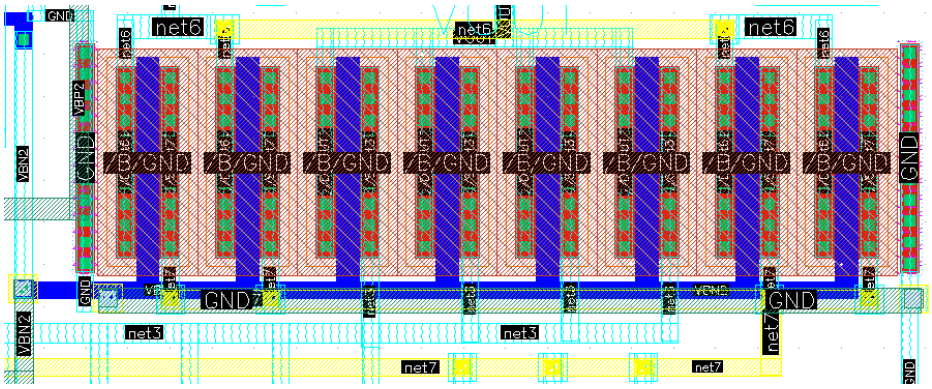


Figure 38: Layout of transistors M5 and M6

5. Transistors M7 and M8

Same case and structure as transistors M11 and M12 , but in this case split in 8 transistors in the structure $YYZZZZYY$. Layout shown in figure 39.

- Transistor M7: Y
- Transistor M8: Z

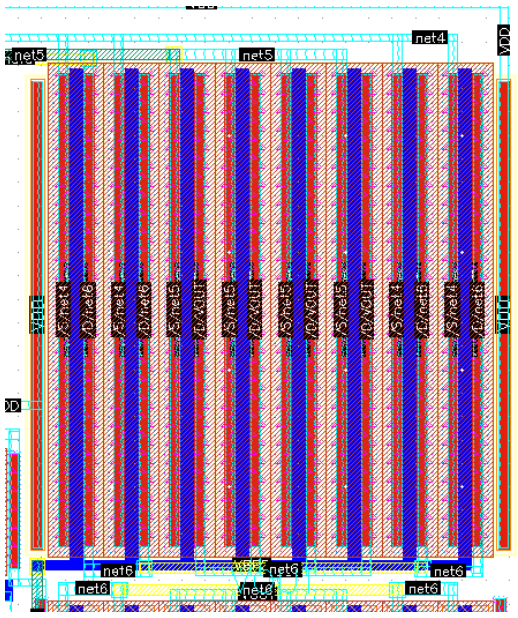


Figure 39: Layout of transistors M7 and M8

6. Transistors M9 and M10

For the layout of this pair of transistors, besides the common-centroid structure, interdigitation is also used because they share a common connection to V_{DD}:

$({}_SY_DY_SZ_DZ_SZ_DZ_SY_DY_S)$

The legend of the transistors is:

- Transistor M9: Y
- Transistor M10: Z

The layout implementation is given in figure 40.

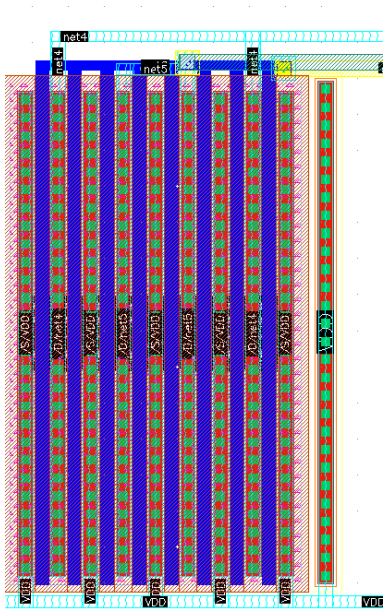


Figure 40: Layout of transistors M9 and M10

Transistor M0 does not require any matching technique, but to reduce the size of it, it is divided in 16 transistors and interdigitation is applied as shown in figure 41. The structure for this transistor is:

$({}_DY_SY_DY_SY_DY_SY_DY_SY_DY_SY_DY_SY_DY_SY_DY_SY_D)$

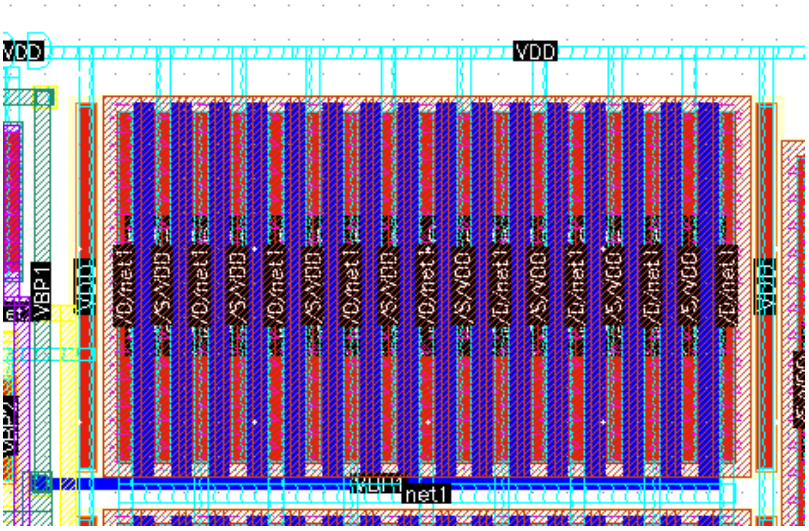


Figure 41: Layout of transistors M0

6.3 Biasing circuit

The transistors in the biasing circuit do not require any common-centroid structure, because it is not a hard-requirement that the transistors are matched, thus, for the transistors in this side of the circuit, only interdigitation is applied in the transistors that allow to apply it to reduce the size of the biasing circuit when doing the layout.

Moreover, another modification made is that all transistors are split thus the size in the layout is reduced.

Below the list of transistors that are grouped by applying interdigitation:

1. Transistors Q7, Q9 and Q11: Common connection to V_{DD}. Layout in figure 42.

2. Transistors Q5 and Q12: Common connection to GND. Layout in figure 43.

3. Transistors Q15 and Q16: Common connection to GND. Layout in figure 44.

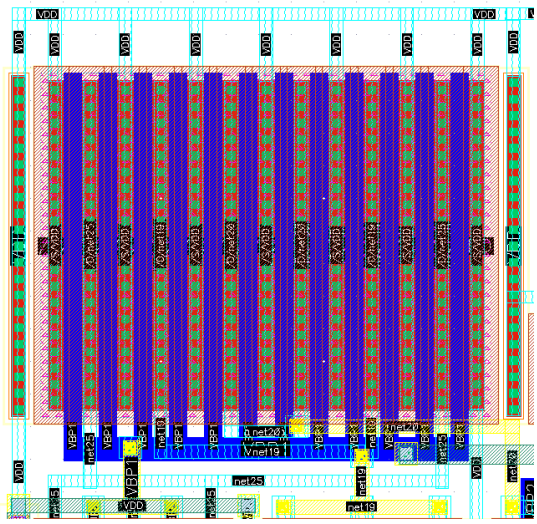
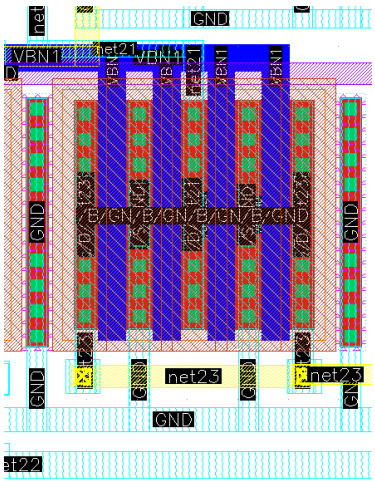


Figure 42: Layout of transistors Q7, Q9 and Q11


net
GND
VBN1
net21
VBN1
VBN1
VBN1

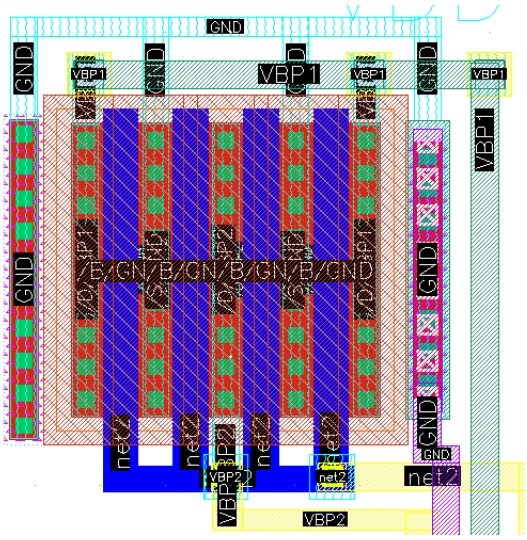


Figure 44: Layout of transistors Q15 and Q16

For the rest of transistors, self-interdigitation is applied, in a similar way that is performed with transistor M0.

The final layout is shown in figure 45.

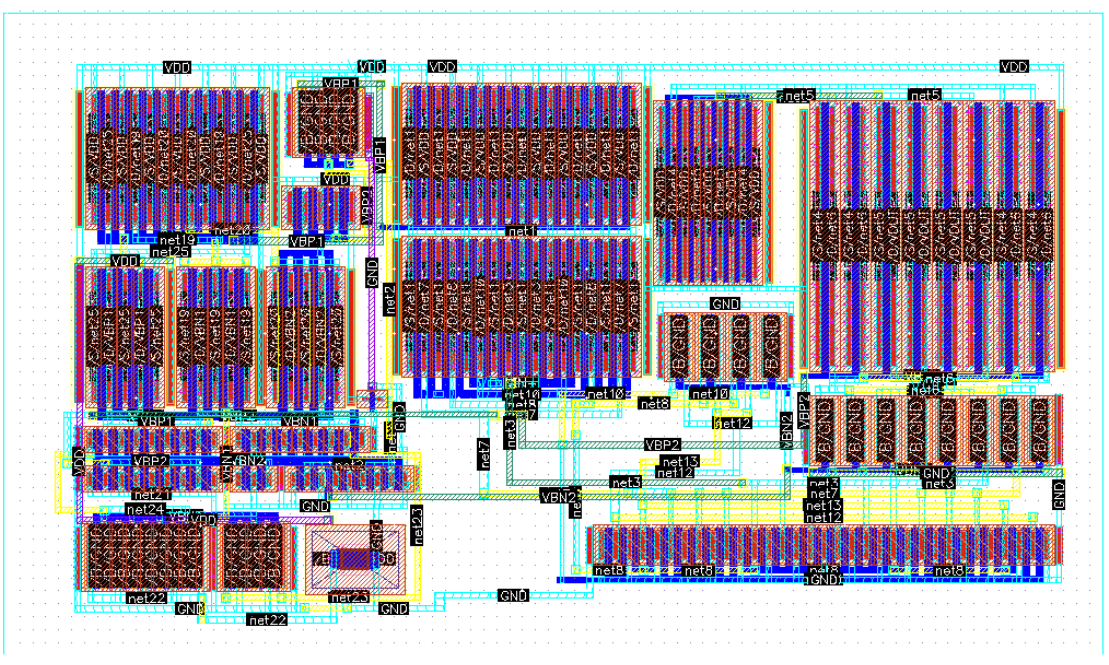


Figure 45: Complete Layout

6.4 Design Rules Check (DRC)

The design showed in figure 45 did not show any Design Rules error, thus, the design was correct from the foundry Process Design Kit (PDK) as shown in figure 46, where no errors appear in the DRC.

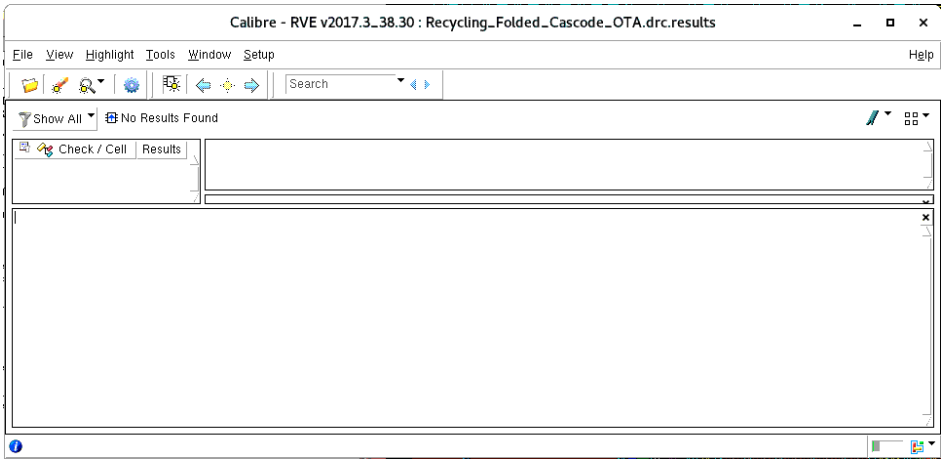


Figure 46: DRC Results

6.5 Layout vs Schematic (LVS)

When it comes to the LVS check, where it is checked that all the components in the schematic are present in the layout, no errors appeared. Results of this check are shown in figure 47.

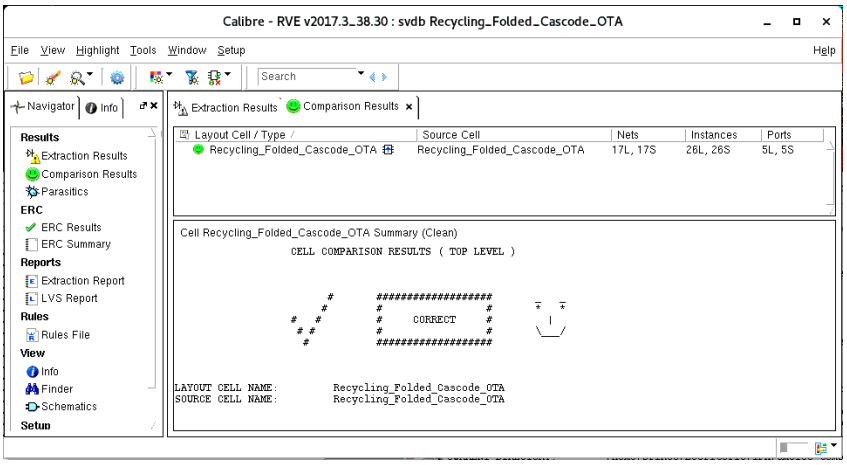


Figure 47: LVS Results

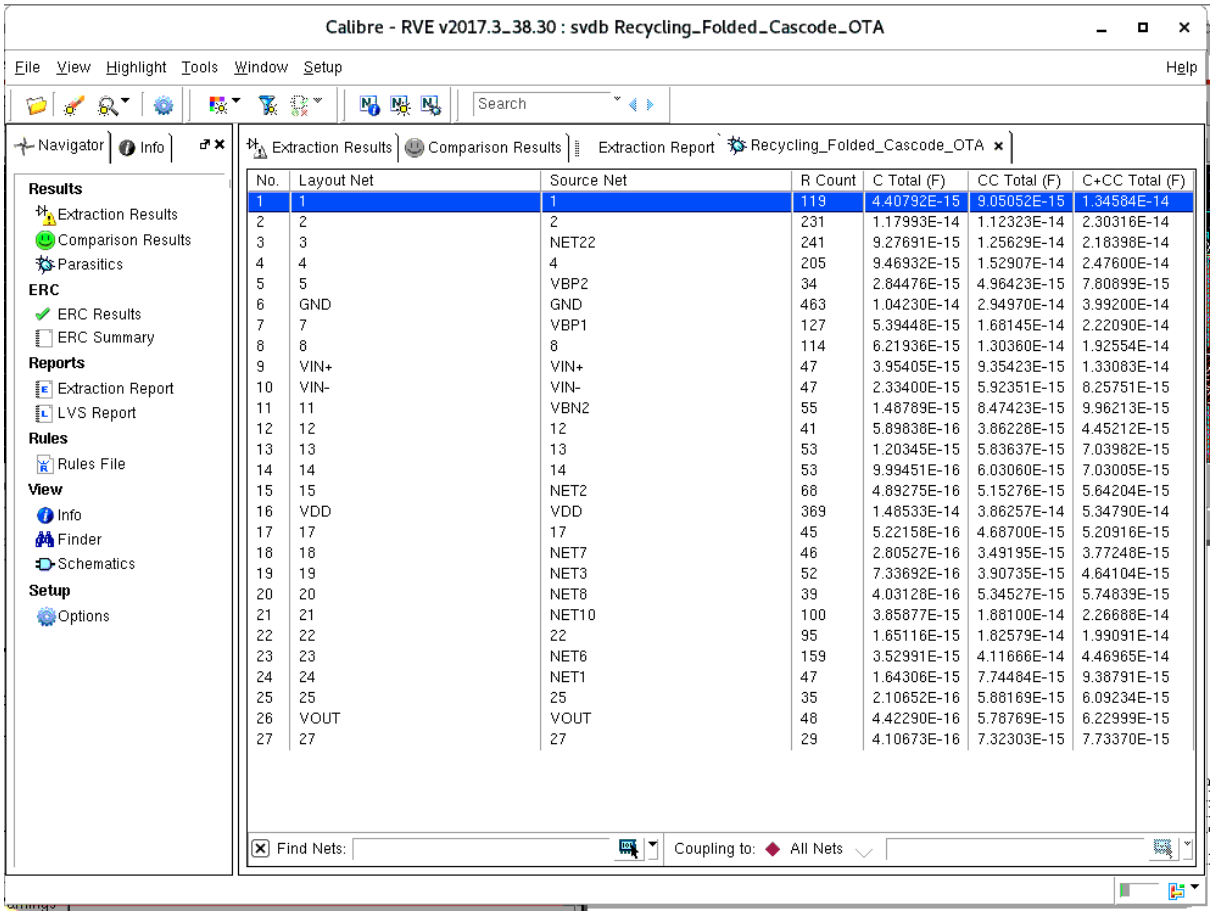
6.6 Parasitic Extraction (PEX)

After the correct LVS check, the next step for the layout validation is the Parasitic Ex-

traction (PEX), which extracts the capacitors, coupled capacitors and parasitic resistors of the design. The output of this check is the closest simulation to the reality.

As mentioned in the introduction, one of the points that lead to delays and discrepancies from the initial Gantt diagram was found at this point. The issues and the solutions found for the different points are presented in this point.

The first issue that was encountered was that the PEX simulation was not outputting the parasitic elements of the circuit. Even after checking the runset that was provided by the Thesis Advisor and all the configuration parameters, it was observed that there was something wrong in the configuration, given that, with the exact same runset and configuration, the PEX on one side was outputting all the parasitics, whereas on the other side it was not. This issue was solved by doing a reinstallation of the complete configuration. After doing this, the issue was solved and the output of the PEX was the expected and all the parasitics were included. The parasitics values are shown in figure 48.



No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	1	1	119	4.40792E-15	9.05052E-15	1.34584E-14
2	2	2	231	1.17993E-14	1.12323E-14	2.30316E-14
3	3	NET22	241	9.27691E-15	1.25629E-14	2.18398E-14
4	4	4	205	9.46932E-15	1.52907E-14	2.47600E-14
5	5	VBP2	34	2.64476E-15	4.96423E-15	7.60899E-15
6	GND	GND	463	1.04230E-14	2.94970E-14	3.99200E-14
7	7	VBP1	127	5.39448E-15	1.68145E-14	2.22090E-14
8	8	8	114	6.21936E-15	1.30360E-14	1.92554E-14
9	VIN+	VIN+	47	3.95405E-15	9.35423E-15	1.33083E-14
10	VIN-	VIN-	47	2.33400E-15	5.92351E-15	8.25751E-15
11	11	VBN2	55	1.48789E-15	8.47423E-15	9.96213E-15
12	12	12	41	5.89838E-16	3.86228E-15	4.45212E-15
13	13	13	53	1.20345E-15	5.83637E-15	7.03982E-15
14	14	14	53	9.99451E-16	6.03060E-15	7.03005E-15
15	15	NET2	68	4.89275E-16	5.15276E-15	5.64204E-15
16	VDD	VDD	369	1.48533E-14	3.86257E-14	5.34790E-14
17	17	17	45	5.22158E-16	4.68700E-15	5.20916E-15
18	18	NET7	46	2.80527E-16	3.49195E-15	3.77248E-15
19	19	NET3	52	7.33692E-16	3.90735E-15	4.64104E-15
20	20	NET8	39	4.03128E-16	5.34527E-15	5.74839E-15
21	21	NET10	100	3.65877E-15	1.88100E-14	2.26688E-14
22	22	22	95	1.65116E-15	1.82579E-14	1.99091E-14
23	23	NET6	159	3.52991E-15	4.11666E-14	4.46965E-14
24	24	NET1	47	1.64306E-15	7.74484E-15	9.38791E-15
25	25	25	35	2.10652E-16	5.88169E-15	6.09234E-15
26	VOUT	VOUT	48	4.42290E-16	5.78769E-15	6.22999E-15
27	27	27	29	4.10673E-16	7.32303E-15	7.73370E-15

Figure 48: Parasitics

After that, a first Post Layout simulation was done. In this case, only the Open Loop and Bandwidth simulations. The results were not expected, given that the GBW was 10 times lower. By looking in the Parasitics section in the Results Viewing Environment (RVE), it was observed that the parasitic elements were not that high to have such reduction in the GBW. By looking closer into the Calibre output of the PEX, it was observed that the resistor from the design was not connected. The resistor was in the Calibre, but there were 3 additional pins besides all the parasitic elements. Those 3 additional pins correspond to the pins of the resistor and those were floating.

The first step to debug this issue was to change the resistor and the layout. This was done to check if the issue was caused by a single resistor. The resistor was changed in the schematic and then the layout was quickly changed to pass DRC and LVS and then, PEX. The layout for this tests was not optimized, it was only done for testing purposes. The conclusion obtained for all those tests was that all the resistors had the same behavior, thus the decision was to keep the first given that it was the layout that was better optimized in terms of size.

The solution found for this issue was to manually connect the resistor. To do so, after running the PEX, when the Calibre View Setup opens, instead of opening in Read Mode, open it in Edit Mode. Then manually do the connection of the resistor to where it corresponds. In this case, the resistor was connected between the source of the transistor of the biasing circuit and ground. The final calibre file output view is shown in figure 49 with the implemented solution in the resistor.

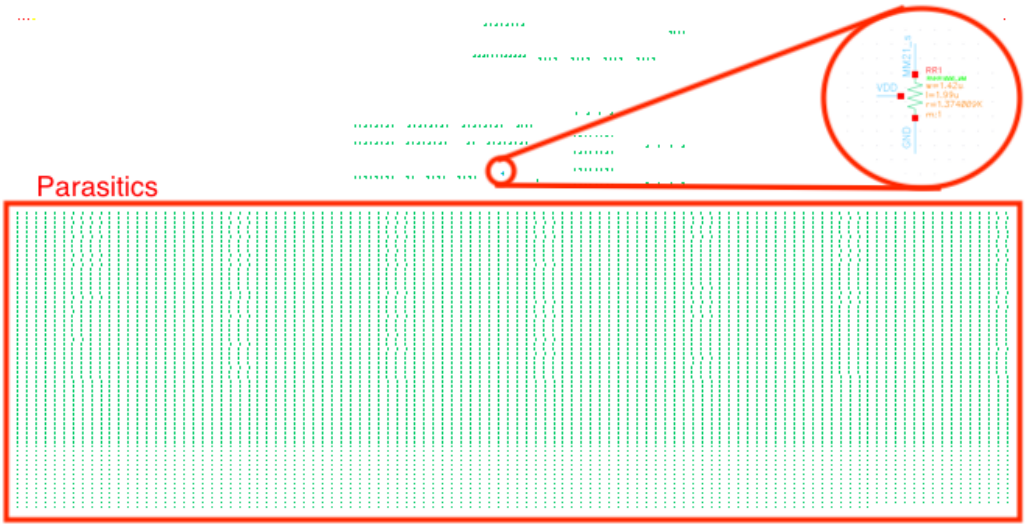


Figure 49: Calibre File with amplified view of the resistor fix

After this connection was done, the Post Layout simulations were done.

7 Post Layout Simulations

In this section, the same circuits and simulation conditions as in the Pre Layout Simulations are performed, thus in the following subsections, only the results comparing Pre Layout Simulations and Post Layout Simulations are shown.

7.1 Open Loop Gain

After running the same circuit and conditions as in the Pre Layout simulations, the obtained results for the amplifier after the layout is given in figure 50. The Open Loop Gain for the Post Layout Recycling structure has a similar value of Gain as the ideal Recycling structure, but it is slightly higher at 58.78dB compared to the previously obtained of 57.91dB for the ideal Recycling structure.

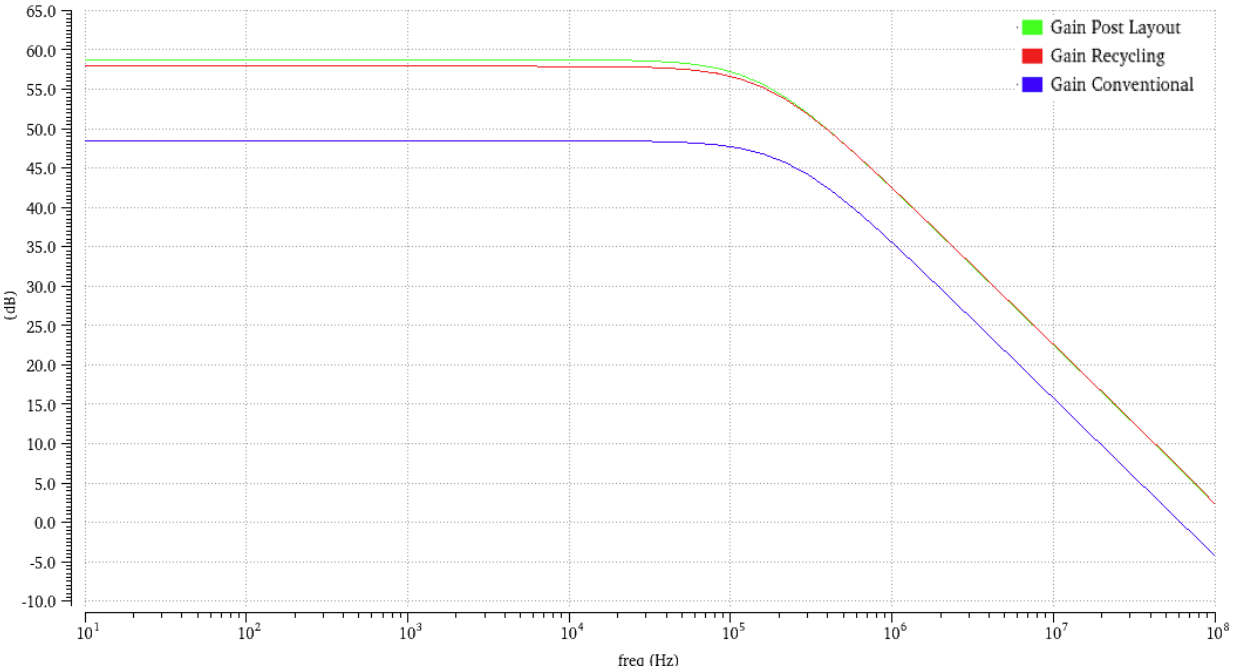


Figure 50: Post Layout Gain Simulation

	Open Loop Gain
Conventional Structure	48.46dB
Current Recycling Structure	57.91dB
Post Layout Current Recycling Structure	58.78dB

Table 18: Open Loop Gain values for the post layout simulations

7.2 Bandwidth

For the bandwidth, the results are shown in table 19. The results for the bandwidth show

that the bandwidth value for the Post Layout is slightly lower than the theoretical value, also, if comparing the GBW from table 20 it is observed that the GBW it is lower. This is explained because there is an increase in the current consumption, which makes increase slightly the gain, as shown before. This increase in current also leads to a decrease in the output resistance, that should lead to an increase in the bandwidth. But due to the effects of the parasitics it makes both bandwidth and GBW decrease.

	Bandwidth
Conventional Structure	233.9 kHz
Current Recycling Structure	173.2 kHz
Post Layout Current Recycling Structure	155.1 kHz

Table 19: Bandwidth values for the post layout simulations

	Gain-Bandwidth Product (GBW)
Conventional Structure	63.66 MHz
Current Recycling Structure	133.2 MHz
Post Layout Current Recycling Structure	132.6 MHz

Table 20: GBW values for the post layout simulations

7.3 Phase Margin

For the stability check, the Recycling structure it is still stable after the layout, as observed in table 21 and the complete AC response in figure 51. Even it is still stable, the reduction in the phase margin must be considered when designing, given the decrease in the margin is roughly 20°.

	Phase Margin
Conventional Structure	82.49°
Current Recycling Structure	62.77°
Post Layout Current Recycling Structure	62.59°

Table 21: Phase Margin for the post layout simulations

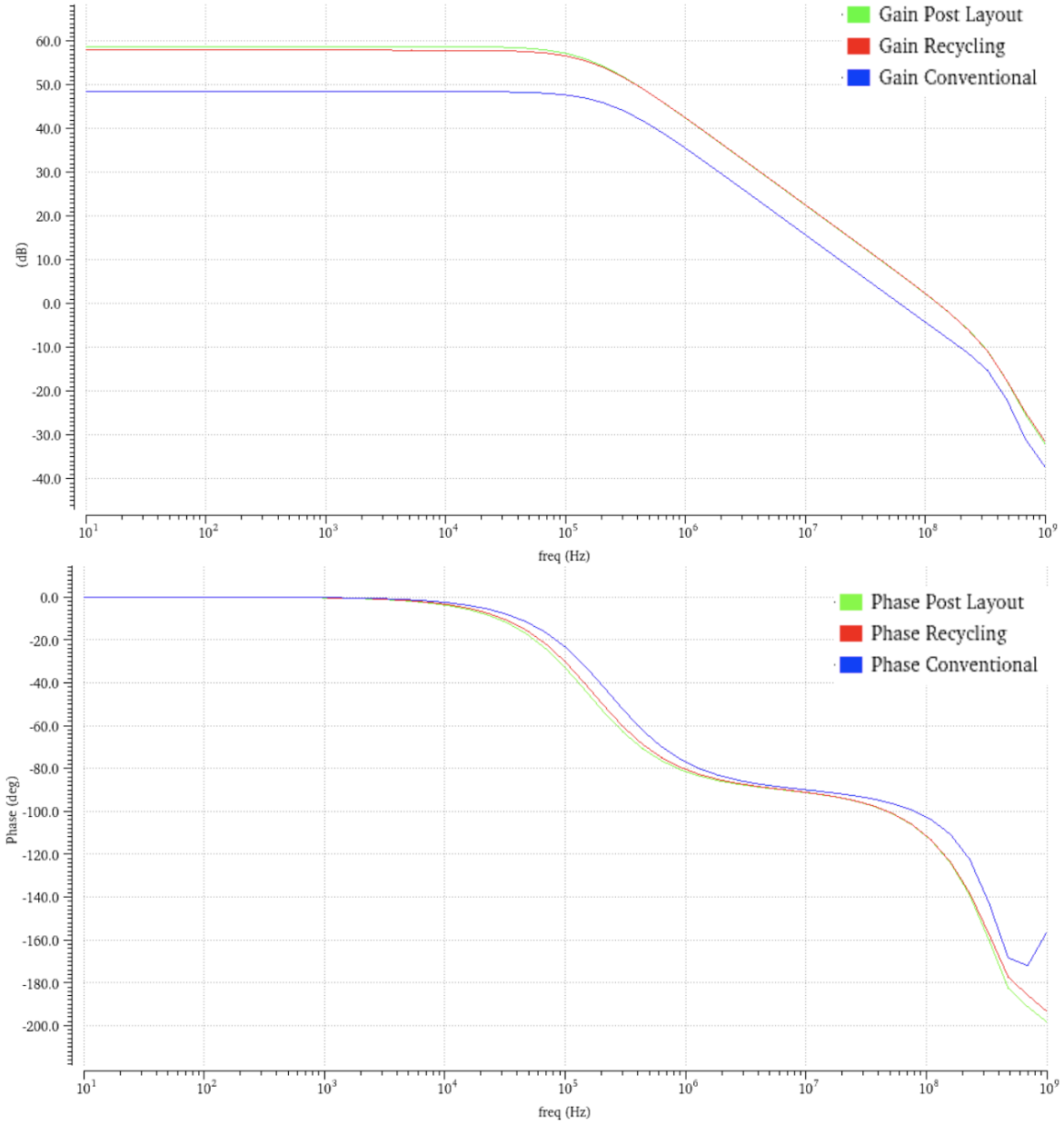


Figure 51: Post Layout complete frequency response

7.4 DC Offset Voltage

The DC Offset Voltage of the three amplifiers is shown in table 22. An observation from this point is that the post layout Recycling structure is the structure that has less DC Offset voltage.

	DC Offset Voltage
Conventional Structure	-942.3μV
Current Recycling Structure	-319μV
Post Layout Current Recycling Structure	-114.3μV

Table 22: DC Offset Voltage for the post layout simulations

7.5 Common Mode Rejection Ratio (CMRR)

With previous obtained offset voltage, the schematic for this simulation and the voltage were changed accordingly and the resulting simulation is in figure 52. It is observed that the CMRR of the Recycling in the Post Layout is higher. This follows the same explanation as explained in the Pre Layout Simulations. Given that the gain of the Post Layout was slightly higher, when dividing by the Common Mode Gain, the expected result is to still be higher than the Pre Layout.

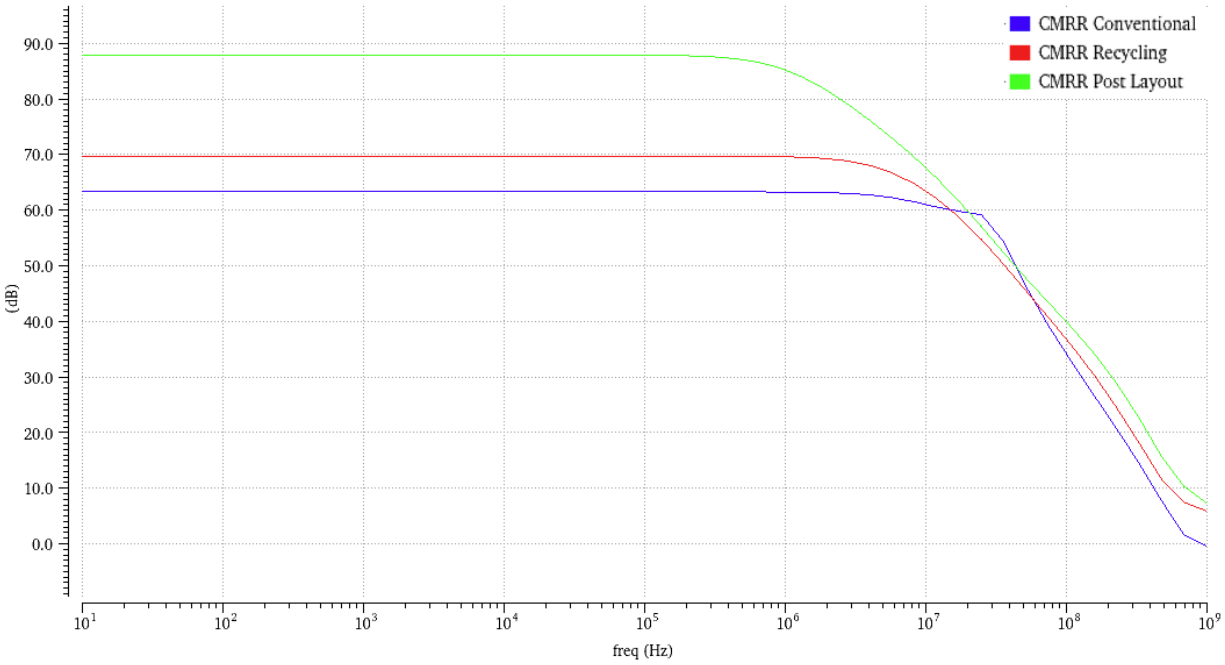


Figure 52: Post Layout CMRR Simulation

	CMRR @ 10Hz	CMRR @ 100MHz
Conventional Structure	63.29dB	34.53dB
Current Recycling Structure	69.72dB	37.06dB
Post Layout Current Recycling Structure	87.89dB	40.16dB

Table 23: CMRR for the post layout simulations

7.6 Power Supply Rejection Ratio (PSRR)

Regarding the PSRR the simulation results is shown in figures 53 and 54 . It is observed that the PSRR is higher due to higher gain in the Post Layout structure because of the previously mentioned increase in the current consumption, that leads to a performance improvement in the PSRR.

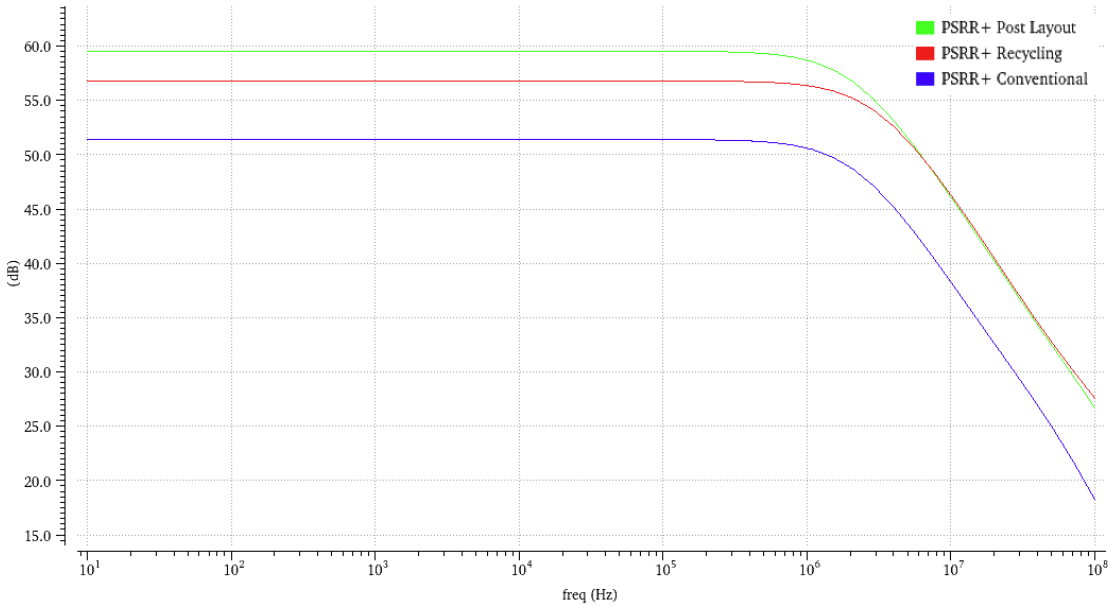


Figure 53: Post Layout PSRR+

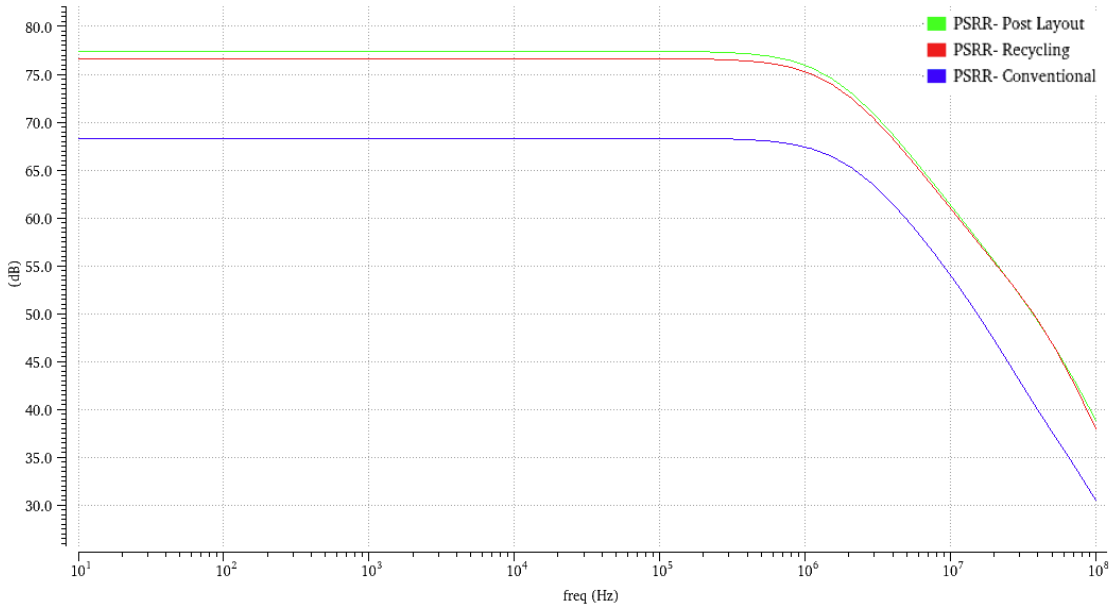


Figure 54: Post Layout PSRR-

	PSRR+ @ 10Hz	PSRR- @ 10Hz
Conventional Structure	51.40dB	68.34dB
Current Recycling Structure	56.79dB	76.65dB
Post Layout Current Recycling Structure	59.54dB	77.42dB

Table 24: PSRR for the post layout simulations

7.7 Input Common Mode Range (ICMR)

No considerable changes in the ICMR after layout. Results in figure 55.

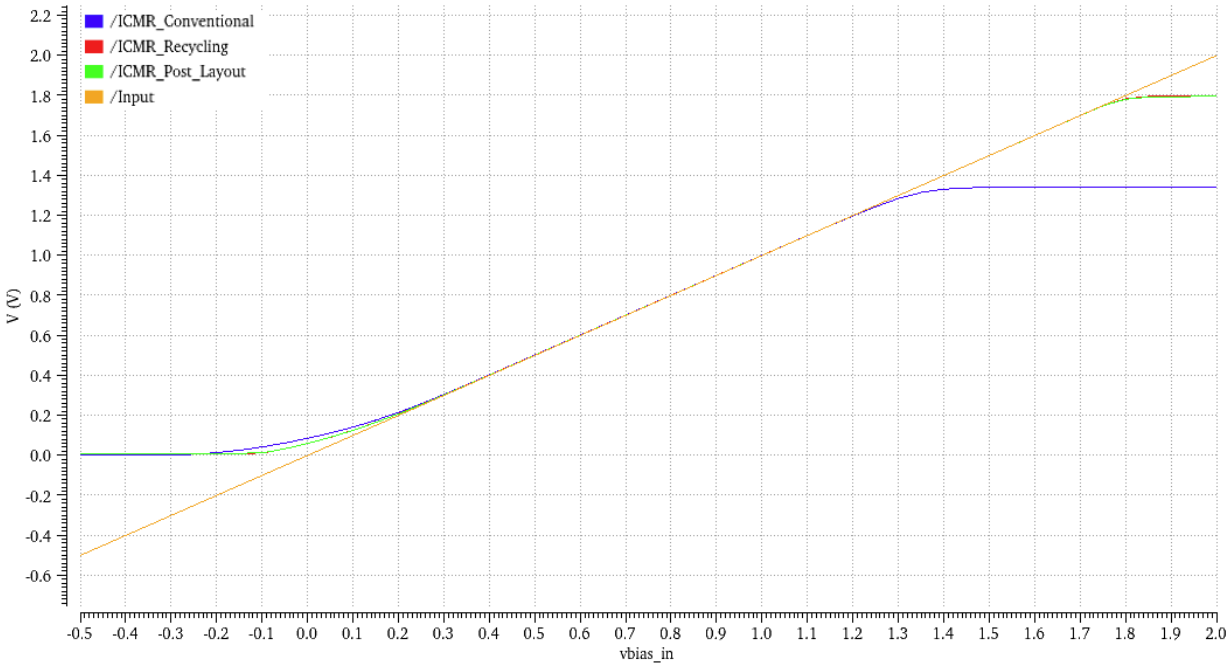


Figure 55: Post layout ICMR

	ICMR [V]
Conventional Structure	[0.3; 1.27]
Current Recycling Structure	[0.3; 1.74]
Post Layout Current Recycling Structure	[0.3; 1.74]

Table 25: ICMR for the post layout simulations

7.8 Slew Rate

As similarly done to previous sections, the transient response, the slew rate visual check

and the values are given in figure 56, 57 and table 26 respectively. From the figures it is observed that no large variations in the Slew Rate after the layout.

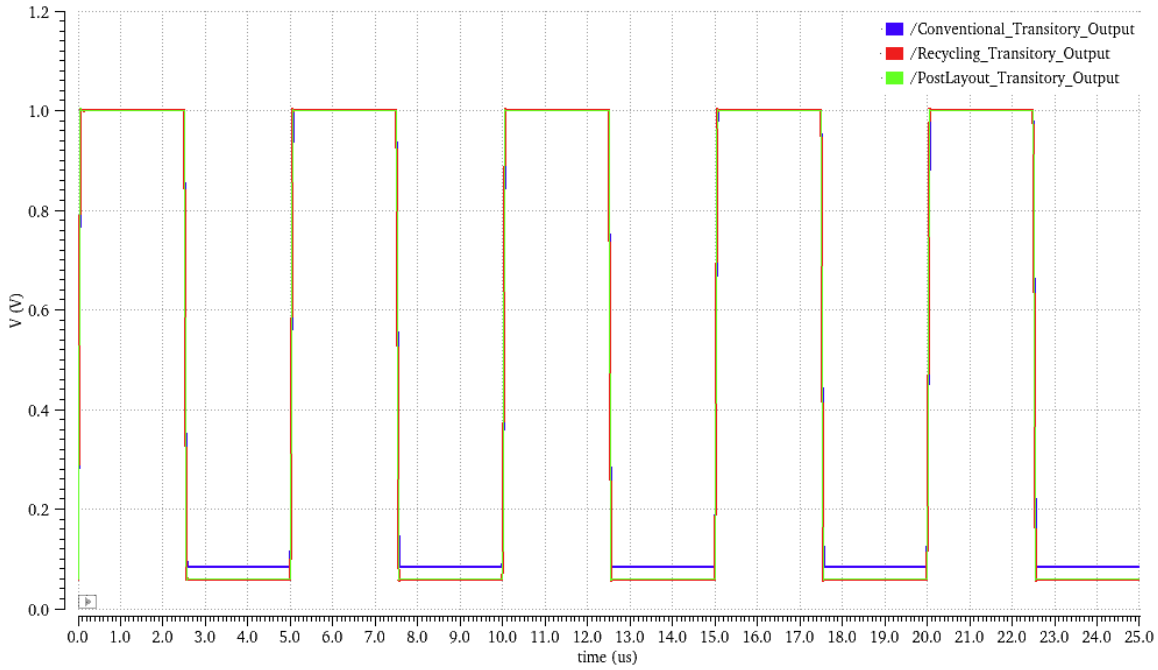


Figure 56: Post layout transient response

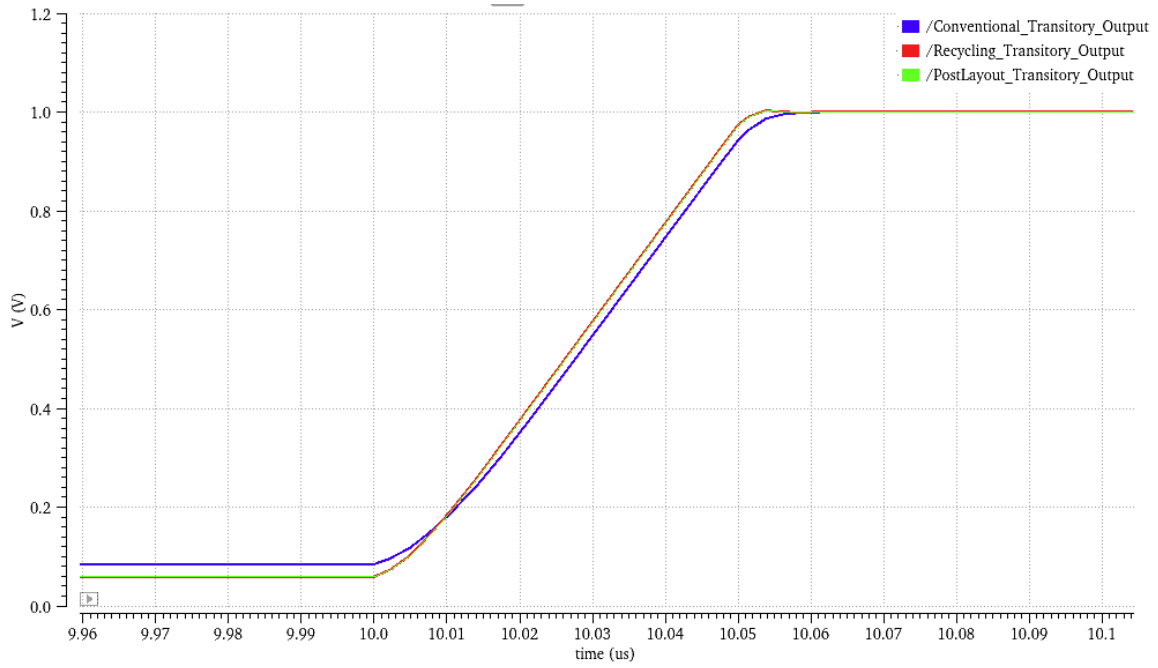


Figure 57: Post layout Slew Rate visual check

	Slew Rate +	Slew Rate -
Conventional Structure	17.69 V/μ s	-18.4 V/μ s
Current Recycling Structure	19.16 V/μ s	-18.95 V/μ s
Post Layout Current Recycling Structure	19.16 V/μ s	-18.94 V/μ s

Table 26: Slew Rate for the post layout simulations

7.9 Settling Time

With the same methodology used, the settling time and overshoot figures are 58 and 59. The obtained values for the settling time and the overshoot do not differ considerably from the theoretical:

	Settling time (90% Final Value)	Overshoot
Conventional Structure	44.67ns	0%
Current Recycling Structure	41.95ns	0.12%
Post Layout Current Recycling Structure	41.98ns	0.10%

Table 27: Settling time and overshoot for the post layout simulations

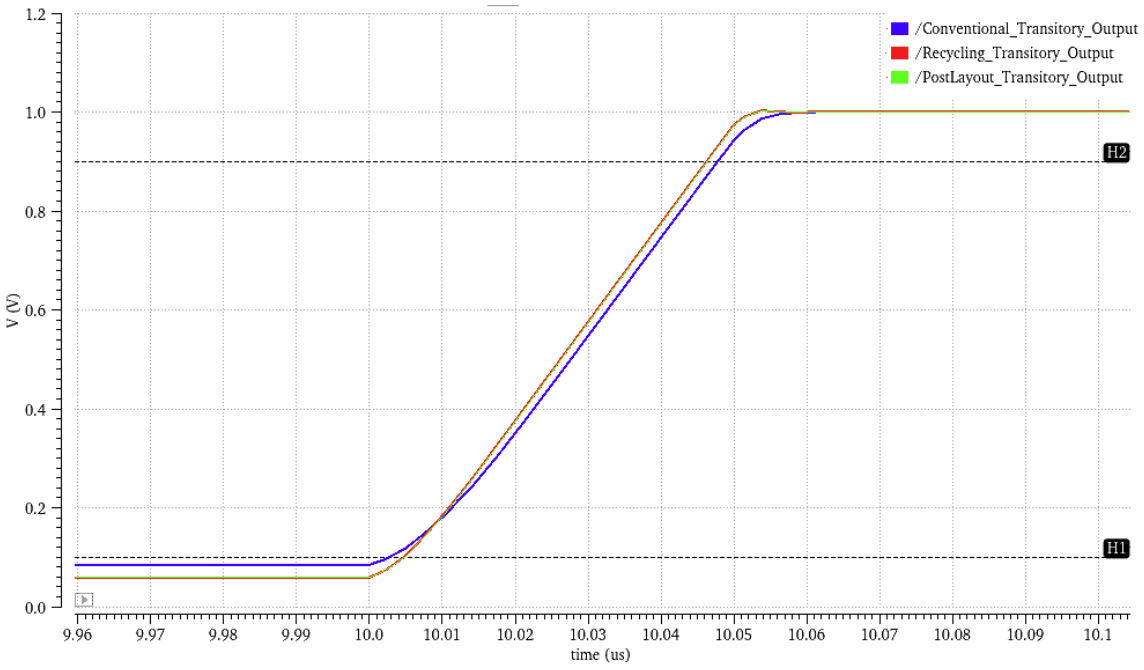


Figure 58: Post Layout settling time markers

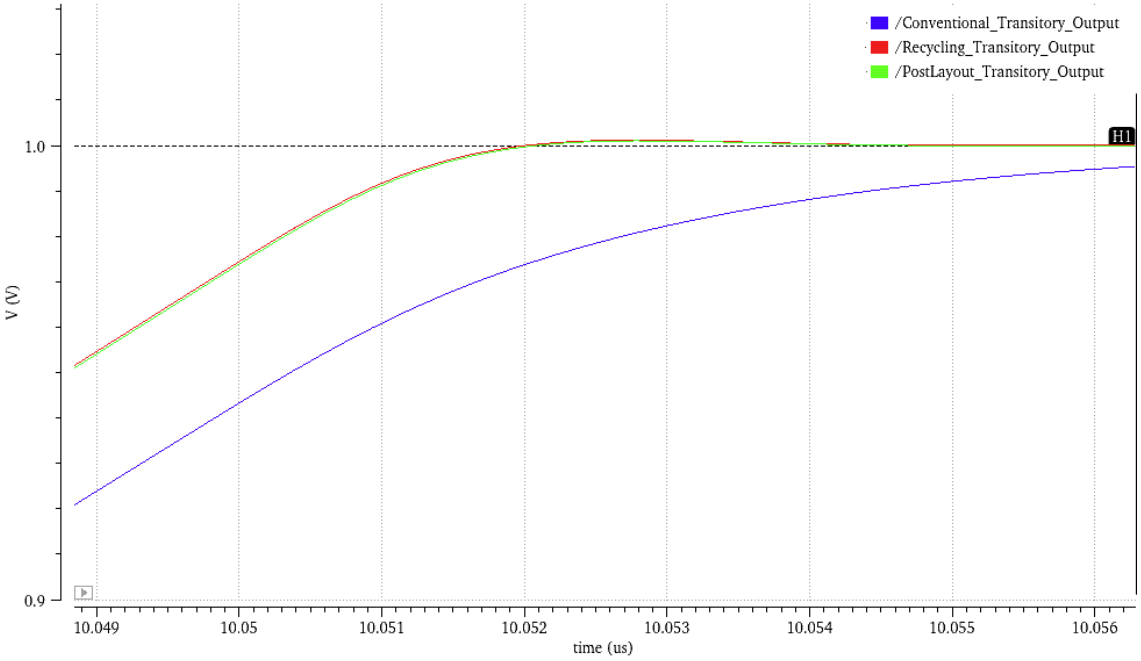


Figure 59: Post Layout close look at the overshoot

7.10 Linearity

The linearity harmonic distortion values are:

	HD2	HD3
Conventional Structure	56.07dB	61.05dB
Current Recycling Structure	55.04dB	59.85dB
Post Layout Current Recycling Structure	55.03dB	59.84dB

Table 28: HD2 and HD3 results for the post layout simulations

The DFT graph is shown in figure 60. In this case, as the values also show, there is no large variation in the results of linearity after the layout.

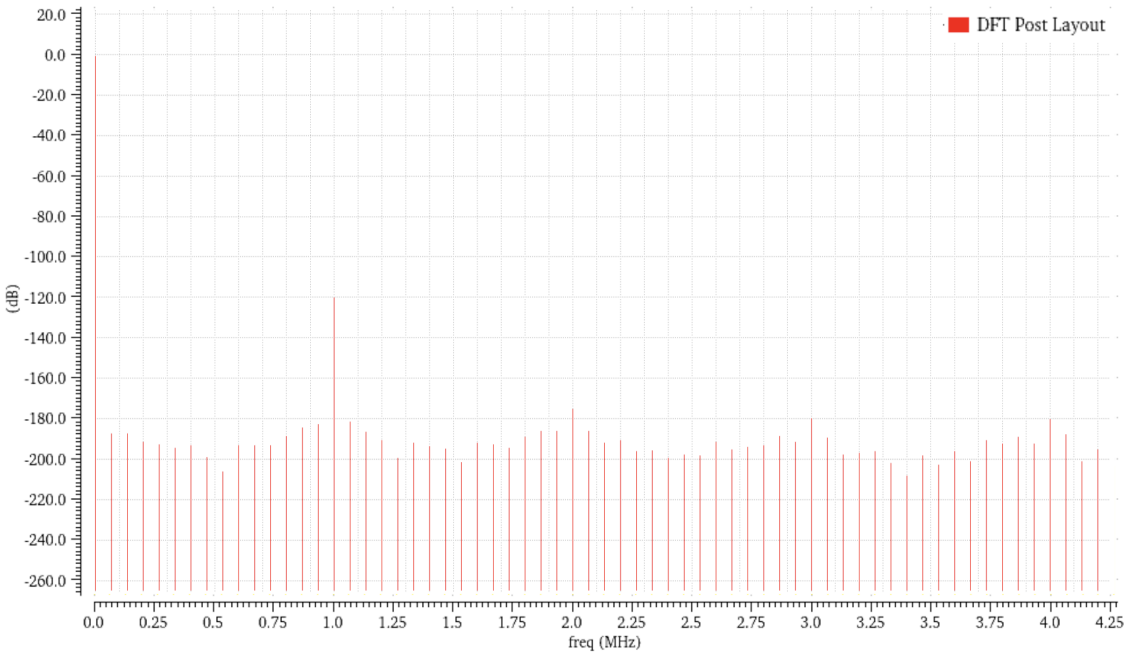


Figure 60: DFT Results for Post Layout Recycling Folded Cascode

7.11 Noise

Noise results are shown in figures 61 and table 29. Results match the theoretical values, where the Recycling structure performs better than the Conventional.

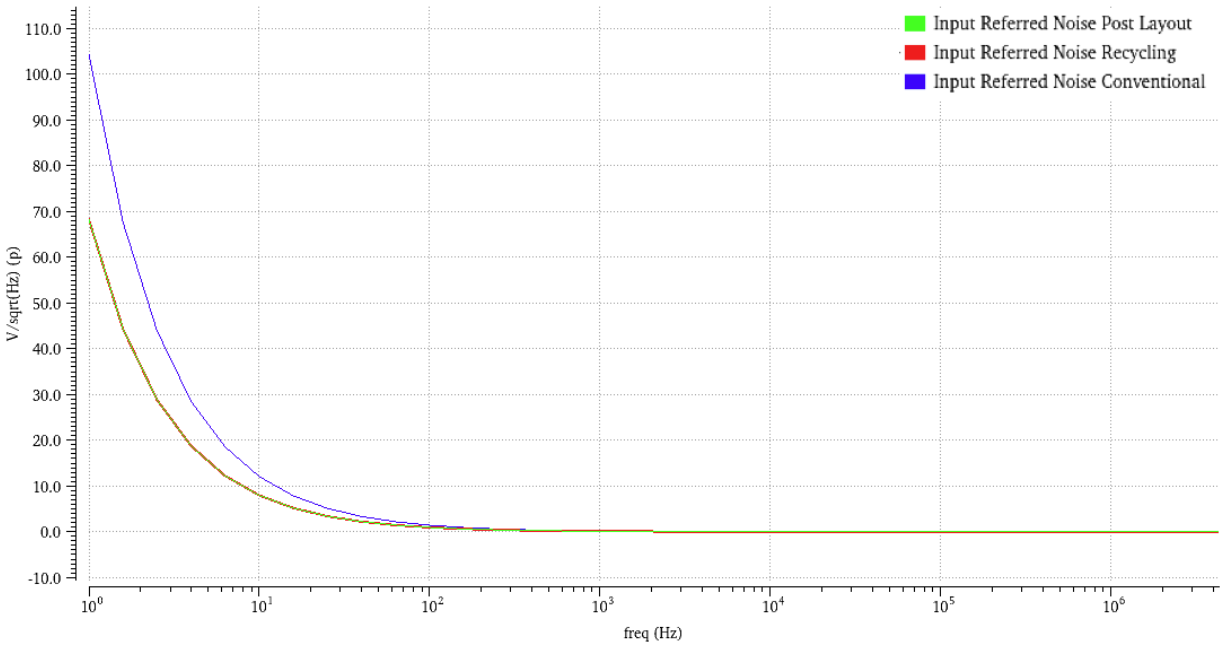


Figure 61: Post Layout Input Referred Noise curve

	Input Referred Noise (Vrms)
Conventional Structure	107μV
Current Recycling Structure	87.45μV
Post Layout Recycling Structure	87.45μV

Table 29: Noise for the post layout simulations

7.12 Discrepances against Pre Layout Simulations

From the previous sections, both Pre Layout Simulations for the Conventional Folded Cascode and the Recycling Folded Cascode are shared, nonetheless, in the table 30 only the Recycling Folded Cascode is compared (pre and post layout).

This simulations only confirm the performance increase shown in previous simulations, given that the performance is similar to what was previously obtained. But there are some parameters that clearly have an unexpected improvement.

Among these results are the Gain, CMRR and PSRR. It is observed that these parameters have an increase of performance, given that the bias current is also increasing, which is not expected at all, because the design has not been modified, so the current consumption of the amplifier should be the same.

The reason of that was checked. Due to the appearance of parasitics, not only in the amplifier, but also in the biasing circuit, the effect on one of the branches of the biasing circuit is affecting the current consumption of the circuit, thus affecting some of the performance. In this case, it leads to an improvement in the performance, mainly due to an increase in the current consumption. As shown in table 30 there is an increase in the bias current of roughly $6\mu A$ after the layout, due to the affectation of the parasitics to the bias circuit.

The other parameters show the expected behavior. Some of them are expected to be slightly lower, given that the effect of the parasitics is usually negative, which is what is observed in general terms in table 30.

Parameter	Pre Layout	Post Layout
Bias Current (μA)	457.2	463.7
DC Gain (dB)	57.91	58.78
Bandwidth (kHz)	173.2	155.1
GBW (MHz)	133.2	132.6
Phase Margin (^{o})	62.77	62.59
Slew Rate + ($V/\mu s$)	19.16	19.16
Slew Rate - ($V/\mu s$)	-18.95	-18.94
Offset Voltage (μV)	-319	-114.3
PSRR+ @ 10Hz (dB)	56.79	59.54
PSRR- @ 10Hz (dB)	76.65	77.42
CMRR @ 10Hz (dB)	69.72	87.89
CMRR @100MHz (dB)	37.06	40.16
ICMR [V]	[0.3; 1.74]	[0.3; 1.74]
Settling time (90% Final Value) (ns)	41.95	41.98
Overshoot	0.12%	0.10%
Input referred noise [$\mu Vrms$] 1Hz - 100Mhz	87.45	87.45
HD2 (dB)	55.04	55.03
HD3 (dB)	59.85	59.84

Table 30: Pre Layout and Post Layout Characterization comparison

8 Economic Study

Since this thesis does not include the tape-out of the design, the cost related to this process is not included in the Economic Study.

The considered wage for the Economic Study is the salary of a Junior Electronic Engineer working full-time during 5 months, since that is the duration of this project. The Social Security contribution is computed separately.

The cost associated for the computer and peripherals is the value that the asset has depreciated during the development of this project.

For the Value-Added Tax (VAT), a 21% of the price is considered for the computer, mouse, keyboard and monitor. For the Cadence license, VPN (Virtual Private Network) and internet the 21% of the associated cost is used and for the electricity a 10% of the associated cost is used.

Resource	**Price**	**Use time**	**Associated Cost**
Junior Engineer Salary	32000€/year	5 months	13333.33€
Social Security contribution	30% of the gross salary	5 months	4000€
Computer	2000€	5 months	500€
Mouse and keyboard	150€	5 months	30€
Monitor	400€	5 months	80€
Cadence license (approximated, not disclosed)	12000€/year	5 months	5000€
VPN (approximated, not disclosed)	10€/month	5 months	50€
Electricity	50€/month	5 months	250€
Internet	40€/month	5 months	200€
Total VAT			1663€
Total			25106.33€

Table 31: Economic Study

9 Conclusions and future development

From previous sections and study presented in this thesis, it is concluded that the recycling structure clearly outperforms the conventional folded cascode operational amplifier. The slight modification with minimal circuit modifications and minimal transistor increase in the circuit that should not complicate excessively the integration, simulation results show that it is a structure worth to switch to if the performance of the conventional structure does not fulfill the expectations.

This circuit should also be preferred when working with tight current budgets, given that provides twice the performance with the same current consumption. Regarding the area, the only difference in terms of size is the addition of two extra transistors and the increase due to the transistor split, that should not increase the area massively.

Nonetheless, as seen during the development of this thesis, as also mentioned in [3], the phase margin decreases in the recycling structure, so that is a point to consider, when switching from a Conventional to a Recycling structure by applying the mentioned changes does not ensure that the system will be stable, mainly because of the decrease in the phase margin due to the addition of a new zero-pole pair.

The stability of the Recycling Structure is sensitive to changes in the load capacitor C_L. As seen in the Pre-Layout Simulations, due to the tight phase margin in the Recycling Structure, slight changes in the load can affect the stability of the design, thus, great care should be taken with this parameter. In the case of the Conventional Structure, since the Phase Margin is larger, the stability of the design does not depend on the load.

The linearity of the Conventional Structure is also slightly better than the Recycling structure.

Some other works show that the increase in the recycling structure keeps improving the performance, which should be interesting to check in future thesis. It has been demonstrated that by recycling the current the performance improves, and from references, with a double recycling technique [15] [6] [12] it is still improving, nonetheless, the performance improvement is not infinite. That study will of course depend on the application, but should be interesting to find the limits until where the recycling structure is really improving and up to what point can this outperform the previous step.

One of the outcomes of this thesis and one of the main gained knowledge is to get familiar with this technology and the provided PDK. Some issues, such as the one presented in the layout section were found. This design does not have a large number of transistors and components, but when the parasitics are added, this number increases. When the circuit complexity increases, the number of impedance that can bring that

issue also increases, thus some internal study must be done to try to find what causes the issue, since in this thesis it was possible to manually fix it because only one resistance was included in the design.

Regarding this thesis, the future steps are the tape-out of the chip and the experimental and on-chip characterization of the design. Many factors must be considered when checking against all the simulations presented in this thesis, such as the value of the load, temperature and testing conditions[3].

Another thesis that could be developed to be compared against this one is to compare two different sizing methodologies and compare which one can lead to a better performance or the advantages and disadvantages that each of the methodologies bring to the design. A methodology that could be used for the sizing is the $\frac{gm}{I_D}$[8].

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